ROYAL GLOBAL UNIVERSITY
GUWAHATI

(ROYAL SCHOOL OF ARCHITECTURE)
(RSA)

SYLLABUS & COURSE STRUCTURE 2026

BACHELOR OF ARCHITECTURE (B.ARCH)

PROGRAMME STRUCTURE
B.ARCH

Total Credits of the Course: 261 Credits								
1st semester B.Arch.								
Sl. No.	**Subject Code**	**Names of subjects**	**L**	**T**	**P/S**	**C**	**TCP**	**SUBJECT TYPE**
PROFESSIONAL CORE COURSES (PC)								
1	ARC132C111	Architectural Design I (Basic Design)	1	0	4(S)	5	5	S
2	ARC132C112	Graphics I	1	0	3(S)	4	4	S
3	ARC132C105	Architecture and Culture	2	0	0	2	2	T
4	ARC132C117	Workshop	0	0	2	2	4	P
5	ARC132M113	Visual Arts	1	0	2	3	5	P
BUILDING SCIENCES AND APPLIED ENGINEERING (BS AND AE)								
6	ARC132C114	Building Construction & Materials I	1	0	4(S)	5	5	S
7	ARC132M106	Structures I	3	0	0	3	3	T
SKILL ENHANCEMENT COURSES								
8	ARC132A111	Communication Skills	2	0	0	2	2	P
TOTAL						**26**	**30**	

2nd semester B.Arch.								
Sl. No.	**Subject Code**	**Names of subjects**	**L**	**T**	**P/S**	**C**	**TCP**	**SUBJECT TYPE**
PROFESSIONAL CORE COURSES (PC)								
1	ARC132C211	Architectural Design II	1	0	7	8	8	S
2	ARC132C212	Graphics II	1	0	3	4	4	S
3	ARC132C204	History of Architecture I	2	0	0	2	2	T
4	ARC132M217	Model Making Workshop	0	0	2	2	4	P
BUILDING SCIENCES AND APPLIED ENGINEERING (BS AND AE)								
5	ARC132C213	Building Construction and Materials II	1	0	4	5	5	S
6	ARC132M205	Structures II	3	0	0	3	3	T
7	ARC132M206	Surveying and Levelling	1	0	1	2	4	T
TOTAL						**26**	**30**	

3rd Semester B.Arch.								
Sl. No.	**Subject Code**	**Names of subjects**	**L**	**T**	**P/S**	**C**	**TCP**	**SUBJECT TYPE**
PROFESSIONAL CORE COURSES (PC)								
1	ARC132C311	Architectural Design III	1	0	7	8	8	S
2	ARC132C305	History of Architecture II	2	0	0	2	2	T
BUILDING SCIENCES AND APPLIED ENGINEERING (BS AND AE)								
3	ARC132C312	Building Construction and Material III	1	0	4	5	5	S
4	ARC132C303	Building Science I	2	0	0	2	2	T
5	ARC132C317	Building Services I	1	0	1	2	3	TP
6	ARC132M306	Structures III	3	0	0	3	3	T
SKILL ENHANCEMENT COURSES								
7	ARC132	Architectural Representation Techniques	1	0	1	2	3	P
8	ARC132S318	Computer Application I	0	0	2	2	4	P
TOTAL						**26**	**30**	

4th semester B.Arch.								
Sl.No.	**Subject Code**	**Names of subjects**	**L**	**T**	**P/S**	**C**	**TCP**	**SUBJECT TYPE**
PROFESSIONAL CORE COURSES (PC)								
1	ARC132C411	Architectural Design IV	1	0	7	8	8	S
2	ARC132C404	History of Architecture III	2	0	0	2	2	T
3	ARC132D413	Landscape Architecture	2	0	1	3	4	TP
BUILDING SCIENCES AND APPLIED ENGINEERING (BS AND AE)								

4	ARC132C412	Building Construction and Material IV	1	0	4	5	5	S
5	ARC132M405	Structures IV	3	0	0	3	3	T
6	ARC132M416	Building Services II	2	0	1	3	4	TP
SKILL ENHANCEMENT COURSES								
8	ARC132S417	Computer Application II	0	0	2	2	4	P
TOTAL						**26**	**30**	

5th semester B.Arch.								
Sl.No.	**Subject Code**	**Names of subjects**	**L**	**T**	**P/S**	**C**	**TCP**	**SUBJECT TYPE**
PROFESSIONAL CORE COURSES (PC)								
1	ARC132C511	Architectural Design V	1	0	7	8	8	S
2	ARC132C503	History of Architecture IV	2	0	0	2	2	T
3	ARC132C508	Theory of Architecture	2	0	0	2	2	T
BUILDING SCIENCES AND APPLIED ENGINEERING (BS AND AE)								
4	ARC132C512	Building Construction and Material V	1	0	4	5	5	S
5	ARC132C514	Building Services III	1	0	2	3	5	TP
6	ARC132G505	Structures V	3	0	0	3	3	T
PROFESSIONAL ELECTIVE (PE)								
7	ARC132D547	Architectural Conservation & Documentation	1	0	2	3	5	TP
TOTAL						**26**	**30**	

Theory of Structures V

6th semester B.Arch.								
Sl.No.	**Subject Code**	**Names of subjects**	**L**	**T**	**P/S**	**C**	**TCP**	**SUBJECT TYPE**
PROFESSIONAL CORE COURSES (PC)								
1	ARC132C611	Architectural Design VI	1	0	8	9	9	S
BUILDING SCIENCES AND APPLIED ENGINEERING (BS AND AE)								
2	ARC132C612	Building Construction and Material VI	1	0	4	5	5	S
3	ARC132C613	Working Drawing	2	0	2	4	6	P
4	ARC132C616	Building Science II	2	0	0	2	2	TP
5	ARC132G604	Universal Design	3	0	0	3	3	T
PROFESSIONAL ELECTIVE (PE)								
6	ARC132D617	Green Buildings/ IKS (Vernacular Architecture)	1	0	2	3	5	TP
TOTAL						**26**	**30**	

7th semester B.Arch.								
Sl.No.	**Subject Code**	**Names of subjects**	**L**	**T**	**P/S**	**C**	**TCP**	**SUBJECT TYPE**
PROFESSIONAL CORE COURSES (PC)								
1	ARC132C711	Architectural Design VII	1	0	8	9	9	S
2	ARC132C702	Town Planning	2	0	0	2	2	T
3	ARC132C713	Urban Design	2	0	1	3	4	TP
BUILDING SCIENCES AND APPLIED ENGINEERING (BS AND AE)								
4	ARC132C704	Disaster Mitigation and Management	1	0	1	2	3	TP
5	ARC132G719	Advanced Construction and Services	1	0	4	5	5	S
PROFESSIONAL ELECTIVE (PE) (Any one)								
6	ARC132D716	Interior Design/ Gender Sensitive Design	1	0	4	5	5	S
PROFESSIONAL ABILITY ENHANCEMENT COMPULSORY COURSES								
7	ARC132C715	Research Methodology	0	0	0	1	2	P
TOTAL						**27**	**30**	

8th semester B.Arch.								
Sl.No.	**Subject Code**	**Names of subjects**	**L**	**T**	**P/S**	**C**	**TCP**	**SUBJECT TYPE**
PROFESSIONAL CORE COURSES (PC)								
1	ARC132C811	Architectural Design VIII	1	0	9	10	10	S
2	ARC132C805	Estimation and Costing	3	0	0	3	3	T
PROFESSIONAL ELECTIVE (PE)								
3	ARC132G807	Contemporary Architecture	3	0	0	3	3	T
PROFESSIONAL ABILITY ENHANCEMENT COMPULSORY COURSES								
4	ARC132C812	Dissertation	0	0	4	4	8	P
5	ARC132C814	Professional Practice and Introduction to Constitutional Law	3	0	0	3	3	T
6	ARC132C806	Construction and Project Management	3	0	0	3	3	T
TOTAL						**26**	**30**	

9th semester B.Arch.								
Sl.No.	**Subject Code**	**Names of subjects**	**L**	**T**	**P/S**	**C**	**TCP**	**SUBJECT TYPE**
PROFESSIONAL ABILITY ENHANCEMENT COMPULSORY COURSES								
1	ARC132C931	Professional Training	NA	NA	NA	26	30	NA
TOTAL						**26**	**30**	

10thsemester B.Arch.								
Sl.No.	**Subject Code**	**Names of subjects**	**L**	**T**	**P/S**	**C**	**TCP**	**SUBJECT TYPE**
PROFESSIONAL CORE COURSES (PC)								
1	ARC132C021	Architectural Thesis	NA	NA	NA	26	30	NA
TOTAL						**26**	**30**	

Scheme of Evaluation of all Architectural subjects shall be as per the norms of The Council of Architecture.

Theory Papers (T): • Continuous Evaluation: Total 50% **(Assignment, Class Test, Viva, Seminar, Portfolio: Any Three)** Mid-term examination: **10%** Attendance: 5% • End Term Examination: 50%	Practical Papers (P): • Continuous Evaluation: 100% **(Skill Test, Jury/ viva, Portfolio, Assignment: Any Three)** Attendance: 5**%**
Combined Theory & Practical Papers (TP):**(To be conducted by the concerned School)** • **Total Assessment -100%** ❖ Continuous Evaluation: 50% Including, 1. (Assignment, Jury, Portfolio, and Viva: Any Three) 2. Mid-term examination: 10% 3. Attendance: 5% ❖ End Semester Theory Examination: 50 %	
Studio Papers (S): with External examiner • Continuous Evaluation: 50% **(Skill Test, Jury/ viva, Portfolio, Assignment: Any Three)** Attendance: 5**%** • End term examination: **50 %**	

1st Semester		
Paper 1	**Architectural Design I (Basic Design)** **L-T-P-C: 1-0-4-5 Credits: 5** **Scheme of Evaluation: Studio**	**Subject Code:** ARC132C111 **Level : 100**

Course Objective: The objective of **Architectural Design I (Basic Design) (ARC132C111)** is to develop an understanding of an existing space used on a regular basis through sketching, measured drawings, and activity analysis.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Identify and understand fundamental design elements such as lines, shapes, forms, textures, and colours in visual composition.	BT1
CO 2	Interpret and apply basic principles of design such as balance, contrast, rhythm, harmony, and unity in 2D and 3D compositions.	BT2
CO 3	Demonstrate proficiency in freehand drawing, sketching, and use of different media for visual representation and design communication.	BT3
CO 4	Analyze and creatively express design ideas through painting and sculpting using various tools and materials.	BT4

Prerequisite:

- NIL

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	Design Elements : • Lines: properties and nature of lines along with usage SHAPE AND FORMS: organic/ geometric shapes and understanding their role w.r.t space (positive/negative).	2	6
II	Texture and colour, Visual scale and weight: • Textures: tactile texture, visual texture and how they are implied • Colour: HUE-SATURATION, VALUE, INTENSITY nature of Colour, implications, Colour wheel, colour scheme. Sense of scale, depth with representation techniques	2	6
III	Design principles: • Balance (symmetrical/asymmetrical) • Contrast (anomaly, emphasis, subordination) • Rhythm/ repetition • Proportion/scale • Harmony and unity. • Variety Radiation	8	24
IV	Freehand Drawing: • Use of various drawing and sketching tools like pencils, ink pens, charcoal pencils, etc.	4	12

	Free hand drawings of household furniture, street furniture, human beings, cars, trees, etc.		
	TOTAL	64	

Remarks:

Design of security cabin, Entrance gate, Memorial, Bus stop etc.

Text Books:

1. *Time saver standards for building types*, De Chiara, Joseph and Crosbie, Michael J.;2nd Ed.; 2011; Tata McGraw Hill; New Delhi.

Reference Books:

1. Neufert, Peter; *Neufert's architects' data;* 4th Ed.; 2012; John Wiley; New Delhi
2. Ramsay and Sleeper; *Architectural graphic standard;* 11th Ed.; 2008; John Wiley; New York.

Reference Materials:

1. Journal

1st Semester		
Paper 2	**Graphics-I** **L-T-P-C: 1-0-3-4 Credits: 4** **Scheme of Evaluation: Studio**	**Subject Code:** ARC132C112 **Level : 100**

Course Objective: The objective of **Graphics-I (ARC132C112)** is to introduce students to the fundamental tools, techniques, and principles of architectural graphics, including freehand sketching, orthographic projections, and presentation tools, enabling accurate and expressive visual communication in architectural design.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Identify and demonstrate the correct use of drawing materials, equipment, and architectural symbols	BT 1
CO 2	Apply principles of orthographic projection and surface development to represent objects accurately	BT 3/BT4
CO 3	Illustrate architectural elements through freehand drawings, plans, elevations, and sections.	BT 3
CO 4	Use tools and graphical techniques effectively for design communication and presentations.	BT 6

Prerequisite:

- NIL

Detailed Syllabus:

Modules	*Topics (if applicable) & Course Contents*	*Periods/ Hours*	
		L	***S***
I	Architectural Graphic Fundamentals–A Lines, lettering and dimensioning, reduction and enlargement of drawings on different scales, representation of materials and architectural elements through architectural graphic symbols	2	6
II	Architectural Graphic Fundamentals –B: Introduction to architectural plans, elevations and sections.	2	6
III	Orthographic Projections: Principles and projection methods of orthographic projection of straight lines, planes, solids. Development of surfaces, sections of solids and intersections of solids.	8	24
IV	Surface Development & Design Tools: Develop proficiency in preparing models using orthographic projection and presentation skills such as power point presentation, graphs designing etc.	4	12
	TOTAL	64	

Text Books:

1. Engineering drawing, Bhatt, N.D and Panchal, V. M;1st Ed.; 2008; Charotar Publishing House; Gujarat.

Reference Books:

1. Ching, F.D.K; Architectural graphics; 6th Ed.; 2015; John Wiley; New York.
2. Smith, Matt; Microsoft office 2010: Ultimate tips and tricks;(e book).

1st Semester		
Paper 3	**Architecture and Culture** **L-T-P-C: 2-0-0-2 Credits: 2** **Scheme of Evaluation: Theory**	**Subject Code:** **ARC132C105** **Level : 100**

Course Objective: The objective of **Architecture and Culture (ARC132C105)** is to familiarize students to the basic introduction of Culture, Society and Early River Valley civilizations. Critical

appreciation of works and synoptic study of architectural characteristic features of various phases and periods.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand the evolution of early human societies and their built environments—including shelters, settlements, and burial systems from the Paleolithic and Neolithic periods.	BT1
CO 2	Analyze the architectural characteristics and urban planning of ancient River Valley civilizations such as Mesopotamia, Egypt, Indus Valley, and Yellow River cultures.	BT2
CO 3	Interpret the architectural elements and philosophies of Classical Buddhism—Mauryan, Hinayana, and Mahayana traditions—focusing on stupas, viharas, and design principles.	BT3
CO 4	Critically examine the architectural features of pre-classical civilizations like Mycenaean, Persian, and Etruscan cultures, drawing comparisons across regions and timelines	BT4

Prerequisite:

nil

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	Pre-Historic World: Paleolithic and Neolithic - Primitive man- Shelters, Settlements, Religious and Burial systems : Oval hut, Nice, Dolmen tomb, Gallery grave, Passage grave, Houses at Catal Huyuk, Stone Henge, Menhirs etc.	8	0
II	River Valley Cultures and Early Civilizations: Mesopotamia, Egypt, Indus Valley, Yellow River. Pyramid of Cheops, Great Ziggurats, Mastabas, Layout of Mohenjo Daro, Harappa, Lothal and Dholavira, Great Bath, Granaries etc.	8	0
III	Classical Buddhism: a) Mauryan and early Buddhist Cultures (Design norms, standards, prescription and style), Philosophies, Viharas , Great Stupa of Sanchi, etc., b) Hinayana & Mahayana Phase.	8	0
IV	Pre-Classical: Mycenean, Persian & Etruscan Civilizations.	8	0
	TOTAL	32	

Text Books:

1. History of architecture in India, Christopher,Tadgell; 2nd Ed.; 2002; Phaidon; London
2. The Great Ages of World Architecture, Hiraskar G. K; 20th Ed.; 2018; Dhanpat Rai & Co

Reference Books:

1. Percy, Brown; Indian architecture, Buddhist and Hindu period; 1st Ed.; 2014; CBS; New Delhi
2. Grover, Satish; Architecture of India: Buddhist and Hindu; 2nd Ed.; 2010; CBS; New Delhi

1st Semester		
Paper 4	**Workshop** **L-T-P-C: 0-0-2-2 Credits: 2** **Scheme of Evaluation: Practical**	**Subject Code:** ARC132C117 **Level : 200**

Course Objective: The objective of **Workshop (ARC132C117)** is to develop skills in understanding various tools, processes and materials.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define basic knowledge of working with structural materials	BT1
CO 2	Discuss the techniques of planning, chiselling, pasting, fixing and welding to learn the use of hand tools.	BT2
CO 3	Discover the use of practical usage of equipment's, processes and its application.	BT3
CO 4	Outline skill in creating designs and making art objects.	BT4

Prerequisite: Nil

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	*P*
I	Carpentry Installation: • Introduction to the carpentry tools, processes, joints and wood working machines • Preparation of various carpentry joints, fixing of plywood, blackboards, commercial boards etc. and their application in furniture. • Handling materials like wood, marble, steel, MS, plywood, POP, Aluminium etc. • Understanding nailing, screwing, riveting and their various conditions and types of applications. • Hands on experience to come up with a product or installation	0	16
II	Foundry Installation: • Introduction, type of patterns, pattern making, preparation of moulds and moulding equipment details. • Hands on experience to come up with a product or installation.	0	16
III	Fabrication Installation: • Introduction to welding equipment, processes and its applications. • Painting and polishing; Classification of paints, varnishes ingredients of paints, painting methods-brush, spray, hot spray etc. • Hands on experience to come up with a product or installation.	0	16
IV	Masonry Installation: • Introduction to basic masonry tools. Understanding various building materials and their tools used for cutting, joining and extension. Hands-on experience to come up with a product or installation.	0	16
	TOTAL	64	

Text Books:

1. Building construction by WB Mckay

Reference Books:

NA

Reference Materials:

NA

1st Semester		
Paper 5	**Visual Arts** **L-T-P-C: 1-0-2-3 Credits: 3** **Scheme of Evaluation: Studio**	**Subject Code:** ARC132M113 **Level : 100**

Course Objective:

The objective of **Visual Arts (ARC132M113)** To develop skills in manual presentation techniques, use of various media of presentation, Principles of 2-D & 3-D compositions and Principles of Design, To help students to understand the visual & aesthetic qualities of Art and its relation to Architectural Design situation, To develop freehand drawing and rendering skills in different medium and using it as tool of expressing ideas visually**.**

Course Objective:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Recall the elements and principle of work of art.	BT1
CO 2	Demonstrate the color techniques to communicate architectural ideas.	BT2
CO 3	Apply own concept and techniques to communicate architectural ideas.	BT3
CO 4	Analyze spatial compositions and visual relationships in two-dimensional and three-dimensional forms.	BT4

Prerequisite: nil

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	Lines: Different kinds of free hand lines, Variations of the lines with pencil, Geometrical object study with pencil shading, Textures with pen & ink	1	11
II	Introduction to Colour/ Colour wheel/ Tint & Shade, Introduction of different techniques of colour, Painting for ancient Heritage.	1	11
III	Geometric sculptures in round (use of medium like clay, plaster of paris, cement wood or terracotta etc.	1	23
IV	Relief sculptures (use of medium like clay, plaster of paris, cement, wood or terracotta etc.)	1	17
	TOTAL	96	

Remarks:

Design of security cabin, Entrance gate, Memorial, Bus stop etc.

Text Books:

1. *Time saver standards for building types*,De Chiara, Joseph and Crosbie, Michael J.;2nd Ed.; 2011; Tata McGraw Hill; New Delhi.

1st Semester		
Paper 6	**Building Construction and Material I** **L-T-P-C: 1-0-4-5 Credits: 5** **Scheme of Evaluation: Studio**	**Subject Code:** ARC132C114 **Level : 100**

Course Objective: The objective of **Building Construction and Material I (ARC132C114)** is to introduce students to the basic materials and techniques used in building construction and to develop foundational knowledge of construction components and methods used in brick and stone masonry.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Identify properties, sources, and applications of basic building materials like sand, lime, stone, and aggregates.	BT1
CO 2	Describe various bonds and construction techniques used in brick and stone masonry.	BT2
CO 3	Apply construction details of foundations, buttresses, lintels, and coping using brick and stone.	BT3
CO 4	Analyze the structural and aesthetic properties of different substitute materials for brick and stone.	BT4

Prerequisite:

- NIL

Detailed Syllabus:

Modules	*Topics (if applicable) & Course Contents*	*Periods/ Hours*	
		L	***P***
I	Introduction to Basic Building Materials: Sand – availability, properties. Aggregate – sources, types. Concrete blocks – types, sizes, qualities, brief on manufacturing. Stone – types, properties, quarrying, aesthetic and structural uses. Lime – types, properties, uses. Lime mortar – preparation, application.	0	37
II	Part A – Bricks and Brick Masonry: Types of brick bonds – Header, Stretcher, English, Flemish, Rat Trap Bond, etc. Part B – Brick Construction Techniques: Foundations, buttresses, lintels, coping.	0	38
III	Part A – Stone Masonry: Types of stone masonry. Part B – Stone Construction Techniques: Foundations, lintels, coping.	0	38
IV	Part A – Brick and Stone Arches: Techniques of constructing brick and stone arches. Part B – Substitute Materials: AAC blocks, Fly ash bricks, Mud blocks (hollow/solid), Glass blocks, Hollow concrete blocks, FRP (Fibre Reinforced Plastic).	0	37
	TOTAL	150	

Text Book:

1. Mackay, W.B.; *Building Construction*, 1st Ed.; 2005; Donheed; London

2. Kumar, S.; *Building Construction*, 20th Ed.; 2010; Standard Publishers.

Reference Books:

1. Chudley, R.; *Construction Technology*, 2nd Ed.; 1987; ELBS; Harlow
2. Barry, R.; *Construction of Buildings*, 4th Ed.; 1999; East West Press; New Delhi

1st Semester		
Paper 7	**Structures I** **L-T-P-C: 3-0-0-3 Credits: 3** **Scheme of Evaluation: Theory**	**Subject Code:** ARC132M106 **Level : 100**

Course Objective: The objective of **Structures I (ARC132M106)** is to introduce the basic principles of structural systems and their behavior under various forces, enabling students to understand why structures stand and how loads are resisted.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Recall the fundamental terms, concepts, and components of structural systems.	BT1
CO 2	Explain various types of structural systems and demonstrate understanding of different types of loads acting on structures.	BT2
CO 3	Apply principles of force, moment, and equilibrium to analyze structural behavior and calculate beam reactions.	BT3
CO 4	Analyze structural sections to determine centroid, moment of inertia, and radius of gyration for basic geometries.	BT4

Prerequisite:

NIL

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	Introduction to the subject and theory of structure: • Technical names and function of various structural components from foundation to roof • Fundamentals and mechanics • S.I. system and units	12	0
II	Structural systems: Understanding loads of various types Load-bearing wall system	12	0
III	Understanding the forces and Moments: • Definition, cause, effect, units • Types of forces, • Conditions of equilibrium • Beam reactions	12	0
IV	Centroid and Moment of Inertia: • Definition, moment of inertia and radius of gyration	12	0
	TOTAL	48	

Text Books:

1. Sanjay; Strength of materials, Bansal, R K and Bansal; 4th Ed.; 2009; Laxmi Publications, New Delhi.
2. S. Ramamrutham – Strength of Materials, Dhanpat Rai Publications, New Delhi
3. S.S. Bhavikatti – Basic Structural Analysis, Vikas Publishing House

Reference Books:

1. Ramamrutham S.; *Strength of materials*; 16th Ed.; 2009; Dhanpat Rai; New Delhi
2. Hibbeler R.C. – Engineering Mechanics: Statics & Dynamics, Pearson Education

2nd Semester		
Paper 1	**Architectural Design II** **L-T-P/S-C –1-0-7-8 Credits: 8** **Scheme of Evaluation: Studio**	**Subject Code:** **ARC 132C211** **Level : 100**

Course Objective: The objective of **Architectural Design II (ARC 132C114)** is to translate knowledge of abstract design principles through enhancement of thought process into a workable design.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**BT Level**
CO 1	Define the characteristics of designing elements in a space and analyze with a design eye and come up with strategic design solutions.	**BT1**
CO 2	Outline the basic design understandings with a sense of aesthetics.	**BT2**
CO 3	Develop an understanding of the importance of presentation drawings.	**BT3**
CO 4	Develop a design solution and representation with the help of collection of user's data, Activity and Area Analysis, Movement and Circulation analysis, Design solution and representation.	**BT4**

Prerequisite:

- Basic Concept of Architecture

Detailed Syllabus:

Modules	*Topics (if applicable) & Course Contents*	*Periods/ Hours*	
		L	***P***
I	**Collection of user's data:** Determining a design problem that could revolve around basic design principles	**6**	**10**
II	**Activity & Area analysis:** Activity & Area analysis of the design problem	**12**	**20**
III	**Movement and circulation analysis:** Movement and circulation analysis of the design problem	**12**	**20**
IV	**Design solution and representation:** Finalization of design and make presentation drawing for execution. Develop sense of material used and construction techniques.	**8**	**40**
	TOTAL	128	

Remarks:

- Design of small residence, doctor's clinic, small cottage, cafeteria etc

Text Book:

1. *Time saver standards for building types*,De Chiara, Joseph and Crosbie, Michael J.;2nd Ed.; 2011; Tata McGraw Hill; New Delhi.
2. Neufert, Peter; *Neufert's architects' data;* 4th Ed.; 2012; John Wiley; New Delhi

Reference Books:

1. Ramsay and Sleeper; *Architectural graphic standard;* 11th Ed.; 2008; John Wiley; New York.
2. Pandya, Yatin; *Elements of space making*; 1st Ed.; 2014; Mapin Publishing; Ahmedabad.
3. Richardson, Phyllis; *XS: Big ideas, small buildings*; 1st Ed.; 2001; Thames & Hudson; London

Reference Materials:

2nd Semester		
Paper 2	**Graphics II** **L-T-P-C: 1-0-3-4 Credits: 4** **Scheme of Evaluation: Studio**	**Subject Code:** ARC132C212 **Level : 100**

Course Objective: The objective of **Graphics II (ARC132C212)** is to develop techniques of sciography, perspective drawing, surface development, etc and to comprehend perspectives of built forms, exercises in parallel, angular and bird's eye views, shades and shadows cast by simple forms on plain surfaces.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define the principles of sciography and identify the behavior of light and shadow on basic geometric and architectural forms	BT1
CO 2	Explain the fundamental components and techniques of perspective drawing including one-point, two-point, and three-point perspectives in architectural contexts.	BT2
CO 3	Demonstrate the process of surface development through isometric and axonometric drawings and apply sciography on perspective views.	BT3
CO 4	Analyse various methods of architectural rendering and evaluate their effectiveness using digital tools like Illustrator and Photoshop.	BT4

Prerequisite:

NIL

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	Sciography: Sciography of Simple Geometric Forms Leading to sciography of hitectural forms.	3	10
II	Perspective Drawing: Definition of perspective technique (picture plane, stationary point etc.) and their role in drawing perspectives, one point, two point and three-point perspectives of geometrical shapes leading to perspectives of built forms, exercises in parallel, angular and bird's eye views, shades and shadows cast by simple forms on plain surfaces.	5	17
III	Surface Development: Development of surfaces of solids, isometric, axonometric of solids. Plotting of sciography on perspective drawings.	3	10
IV	Presentation Illustrator and Photoshop: Graphical presentation and rendering in pen and ink of architectural drawings and materials.	5	27
	TOTAL	80	

Text Books:

Engineering drawing, Bhatt, N.D and Panchal, V. M;1st Ed.; 2008; Charotar Publishing House; Gujrat.

Reference Books:

Ching, F.D.K; *Architectural graphics;* 6th Ed.; 2015; John Wiley; New York

Carpo, Mario and Lemerle, Frederique; *Perspective, projections and design: Technologies of architectural representation*; 2nd Ed.; 2013; Taylor & Francis; Hoboken.

Reference Materials:

Sketching for Architecture + Interior Design by Stephanie Travis

Visualizing Architecture by Alex Hogrefe (Website)

2nd Semester		
Paper 3	**History of Architecture-I** **L-T-P-C: 2-0-0-2 Credits: 2** **Scheme of Evaluation: Theory**	**Subject Code:** ARC132C204 **Level : 100**

CourseObjective: The objective of **History of Architecture-I (ARC 132C204)** is to familiarize students with the development of culture, society, and worldviews of early civilizations through a critical and synoptic study of architectural characteristics across various historical phases and periods.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Recall and describe the early development of Indian art and architecture and its influence on culture and society.	BT1
CO 2	Understand and explain the design principles, spatial composition, and visual characteristics of Classical Greek and Roman architecture.	BT2
CO 3	Apply architectural vocabulary to analyze ecclesiastical structures from Early Christian, Byzantine, Medieval, and Gothic periods.	BT3
CO 4	Examine and critically evaluate the planning and spatial concepts of Vedic architecture, including settlement patterns and Vaastu Shastra principles	BT4

Prerequisite:

NIL

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	History of Indian Art & Architecture: A brief about history of history of Indian Art & Architecture and its influence on Culture and Society.	8	0
II	Classical Greece and Rome: Study of principles of design, proportion, composition, visual effects. Eg. Doric, Ionic, Corinthian, Tuscan & Composite orders, Optical Corrections, Parthenon, Theatre at Epidaurus, Pantheon, Colosseum, Thermae of Caracalla, Pont due gard, Nimes, Basilica of Trajan, Arch of Septimius Severus, Column of Trajan.	8	0
III	Ecclesiastical: Early Christian, Byzantine, Medieval and Gothic Vedic Culture, Kingship in India, Hellenistic influences.	8	0
IV	Vedic Culture: a) Settlement Patterns during Vedic Civilization. b) Vaastu Shastra and its Principles, Layout Planning (Residence) using Vaastu Principles.	8	0
	TOTAL	32	

Text Books:

1. *History of architecture in India*,Christopher, Tadgell; 2nd Ed.; 2002; Phaidon; London.
2. *Indian architecture, Buddhist and Hindu period*,Percy Brown; 1st Ed.; 2014; CBS; New Delhi.

Reference Books:

1. Grover, Satish; *Architecture of India: Buddhist and Hindu*; 2nd Ed.; 2010; CBS; New Delhi

2nd Semester		
Paper 4	**Model Making Workshop** **L-T-P-C: 0-0-2-2 Credits: 2** **Scheme of Evaluation: Practical**	**Subject Code:** ARC132M217 **Level : 100**

Course Objective: The objective of **Model Making (ARC132M217)** is to develop model making skills and visualization.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Recall the tools, materials, and basic techniques used in architectural model making.	BT1
CO 2	Understand and interpret design drawings to create accurate scaled models using a single material.	BT2
CO 3	Apply model making skills to construct complex models using composite materials and multiple elements	BT3
CO 4	Analyze and design detailed site models incorporating landscaping, contours, and site features.	BT4

Prerequisite: NIL

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	Single material model: Paper scaled model	0	12
II	Monochrome model: Scaled model	0	12
III	Composite material model: Scaled model	0	12
IV	Site model with landscaping and site features and contours: Scaled model	0	12
	TOTAL	48	

Text Books:

NA

Reference Books:

NA

2nd Semester		
Paper 5	**Building Construction & Materials II** **L-T-P-C: 1-0-4-5 Credits: 5** **Scheme of Evaluation: Studio**	**Subject Code:** ARC132C213 **Level : 100**

Course Objective: The objective of **Building Construction & Materials II (ARC132C213)** is to introduce students to the technical aspects of building construction and materials, and to acquaint them with various traditional and modern methods of construction.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Recall the types, properties, defects, and applications of timber and commercial wood products in construction	BT1
CO 2	Understand the design, types, and joinery techniques of wooden doors used in buildings	BT2
CO 3	Apply appropriate construction methods and joinery details in designing various types of wooden windows and ventilators.	BT3
CO 4	Analyze the components and structural systems of timber and steel trussed roofs and evaluate the suitability of different roofing materials.	BT4

Prerequisite: NIL

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	Timber and Commercial Wood Quality of timber used in building, defects, seasoning and preservation of timber, types Natural, hard and softwood, uses of timber for aesthetic & structural purposes Uses of commercial wood in building i.e., plywood, block boards, particleboards, veneers and laminates and other types.	0	16
II	Doors: Types of wooden Doors, i.e., ledged, braced, battened, panelled, flush and glazed doors, study of joinery details.	0	16
III	Windows: Types of wooden glazed windows, Fixed, side and top hung pivoted, louvered, ventilators and fanlights, study of joinery details.	0	16
IV	Roof: Timber and steel trussed roof, various parts, their purposes and method of construction. Use of AC sheet, GI sheets and aluminium sheets for roofing.	0	16
	TOTAL	64	

Text Books:

1. *Building construction*;Mackay, W.B. ;*Buildingconstruction*; 1st Ed.; 2005; Donheed; London.
2. *Building Construction*, Kumar S; 20th Ed.;2010;Standard Publishers.

Reference Books:

1. Chudley, R; *Construction technology*; 2nd Ed.; 1987; ELBS; Harlow
2. Barry, R; *Construction of building*; 4th Ed.; 1999; East West Press; New Delhi

2nd Semester		
Paper 6	**Structures II** **L-T-P-C: 3-0-0-3 Credits: 3** **Scheme of Evaluation: Theory**	**Subject Code:** ARC132M205 **Level : 100**

Course Objective: The objective of **Structures II (ARC132M205)** is to introduce students to the fundamental behavior of structural members under various loads, focusing on stress-strain relationships, shear and bending in beams, and the analysis of trusses. The course aims to develop an understanding of how structural systems deform and carry loads, preparing students to apply these principles in architectural and structural design.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Recall and define basic concepts of stress, strain, modulus of elasticity, and material behavior under axial loads.	BT1
CO 2	Explain the relationships between various elastic constants such as modulus of rigidity, Poisson's ratio, and bulk modulus, and interpret their role in material deformation.	BT2
CO 3	Apply concepts of shear force and bending moment to draw diagrams for simply supported and cantilever beams under various loading conditions.	BT3
CO 4	Analyze pin-jointed trusses using the method of joints and method of sections for different loading conditions.	BT4

Prerequisite:

Completion of "Theory of Structures I and Mathematical Tools" is required to understand the fundamental concepts of structural components, force systems, and basic mechanics.

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I (Theory)	**Simple Stresses and Strains:** Concept of Deformable Bodies. Types of stresses (compressive, tensile, shear) and strain (axial, shear, volumetric) Simple problems. Modulus of Elasticity, typical stress strain behaviour of steel and concrete.	12	0
II (Theory)	**Elastic constants:** Elastic constants, Rigidity Modulus, Poisson's Ratio. Bulk Modulus and Shear Modulus, Relations, Modulus of Elasticity and Modulus of Rigidity	12	0
III (Theory)	**Shear and Bending Moments in beams** B.M & S.F diagrams for simply supported and cantilever beams.	12	0
IV (Theory)	**Analysis of Trusses:** Application of Trusses, Composition of forces in a plane, Method of joints, Method of sections	12	0
	TOTAL	48	

Text Books:

1. Sanjay;Strength of materials, Bansal, R K and Bansal; 4th Ed.; 2009; Laxmi Publications, New Delhi.
2. S. Ramamrutham – Strength of Materials, Dhanpat Rai Publications, New Delhi
3. S.S. Bhavikatti – Basic Structural Analysis, Vikas Publishing House

Reference Books:

1. Ramamrutham S.; *Strength of materials*; 16th Ed.; 2009; Dhanpat Rai; New Delhi

2. Hibbeler R.C. – Engineering Mechanics: Statics & Dynamics, Pearson Education
3. M.L. Gambhir – *Fundamentals of Structural Mechanics and Analysis*, PHI Learning

2nd Semester		
Paper 7	**Surveying and Levelling** **L-T-P-C: 1-0-1-2 Credits: 2** **Scheme of Evaluation: Theory**	**Subject Code:** ARC132M206 **Level : 100**

Course Objective: The objective of **Surveying and Levelling (ARC132M206)** is to introduce students to the fundamental principles and methods of surveying and levelling essential for architectural and construction practices. It aims to develop a thorough understanding of measurement techniques, various surveying tools, and the ability to prepare and interpret maps, contour plans, and layout drawings for site development and planning.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define the fundamental principles and classifications of land surveying and measurement techniques.	BT1
CO 2	Explain and distinguish between different types of plane surveying methods including chain, compass, and plane table surveys.	BT2
CO 3	Apply levelling and theodolite surveying techniques for computation of reduced levels (RL), adjustments, and identification of errors.	BT3
CO 4	Analyze and interpret contour maps; calculate areas and volumes; and execute layout of site features like building centerlines, sewers, and roads using contour and setting-out methods.	BT4

Prerequisite: NIL

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	**Introduction:** Principles & Classification of survey, Basic measurements, Principles and methods of surveying	16	0
II	**Plane Surveying:** Chain survey, Compass survey, Plane table surveying	16	0
III	**Theodolite &Levelling:** Theodolite survey, Types of levelling, sources of errors, Computations & Permanent adjustment of levels, Problems on RL	16	0
IV	**Contouring and Application:** Characteristics of contours, direct and indirect methods of contouring, interpolation, uses of contours, setting out works such as centre lines of a building, grade for sewer, centre line of a bridge. Area and Volume calculation.	16	0
	TOTAL	64	

Text Books:

1. Surveying Vol I" by PC Punmia.

Reference Books:

1. Kanetkar TP and Kulkarni SV; Surveying and Levelling (Part-I)"
2. N.N. Basak; Surveying and Levelling

3rd Semester		
Paper 1	**Architectural Design III** **L-T-P-C: 1-0-7-8 Credits: 8** **Scheme of Evaluation: Studio**	**Subject Code:** ARC132C311 **Level : 200**

Course Objective: The objective of **Architectural Design III (ARC132C311)** to expose students to vernacular architecture in terms of local construction technique, building material, architectural language of a particular region, to understand the impact of socio-cultural, climate, topography etc. of the region in architecture of a place, to make attempts in designing religious precincts, community centres, vocational training institute, primary schools etc.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Recall and identify the socio-cultural, climatic, and topographical influences on the vernacular architecture of a region through basic study and statistical mapping.	BT1
CO 2	Understand and interpret the local construction techniques, materials, and architectural language through documentation and case studies of regional architecture	BT2
CO 3	Develop spatial concepts and planning strategies for community-based architectural projects such as schools or health centers using contextual design approaches.	BT3
CO 4	Analyze, finalize, and effectively represent architectural design solutions through detailed site planning, drawings, and portfolio presentations.	BT4

Prerequisite: NIL

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	**Study and mapping:** Statistical analysis of the region to determine various user groups to establish a relationship between human feelings and the built form.	1	31
II	**Study and documentation of local style of architecture and material used and derive concepts:** Use of locally available resources and its implication on the economy of the built environment.	1	31
III	**Concept Development and planning/zoning:** To develop clusters and community formations through space planning. Discussion and development of design	1	31
IV	**Design solution and representation:** Finalization of design and make presentation drawing for execution. Presentation of works of design through portfolio or panels	1	31
	TOTAL	128	

Text Books:

1. Time saver standards for building types; De Chiara, Joseph and Crosbie, Michael J.; Time saver standards for building types; 2nd Ed.; 2011; Tata McGraw Hill; New Delhi.

Reference Books:

1. Christopher, Alexander; The Timeless Way of Building; 1st Ed.; 1979; Library of Congress Cataloguing in Publication Data; London.
2. Bhatia, Gautam; Lauri Baker: Life, Works and Writings; 1st Ed.; 1994, Penguin Books India Limited; New Delhi.

3rd Semester		
Paper 2	**History of Architecture II** **L-T-P/S-C: 2-0-0-2 Credits: 02** **Scheme of Evaluation: Theory**	**Subject Code:** ARC132C305 **Level : 200**

Course Objective: The objective of **History of Architecture II** (ARC132C305) is to provide an understanding of the evolution of Hindu Architecture in India in its various stylistic masses, characterized by technology, ornamentation and planning practices

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Describe the evolution and typologies of Hindu temple architecture, including early examples from Udaigiri, Tigawa, and Aihole.	BT1
CO 2	Explain the distinguishing features of Dravidian and Indo-Aryan styles through analysis of architectural examples from Pallava, Chola, Pandya, and Hoysala periods.	BT2
CO 3	Compare and interpret the regional variations and contributions to Hindu temple architecture across different dynasties and locations such as Orissa, Gujarat, and Khajuraho.	BT3
CO 4	Evaluate the cultural and architectural significance of major temple complexes from the Later Dravidian period, including those at Hampi, Madurai, and Srirangam.	BT4

Prerequisite: NIL

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/Hours***
		L
I	**Evolution of Hindu temple:** Evolution of Hindu temple –both Indo Aryan and Dravidian- Early temples at Udaigiri.Tigawa and Sanchi –Experiments at Aihole (Durga temple and Ladkhan Temple), Deogarh, Bhitargaon and Badami. Beginning of Dravidian architecture – Pallavas, rathas at Mamallapuram, Shore temple, Kailsanatha and Vaikuntaperunal temple at Kancheepuram.	8
II	**The Chola and Pandya:** The Cholas – Brihadeshwara temple at Thanjavur and Gangaikonda Cholapuram The Pandyan contribution- gopurams The Hoysala temples at Belur, Halebid and Somnathpur. Eg.Channakesava Temple, Belur, Hoysalesvara temple, HalebidKesava temple, Somnathpur.	8
III	**Indo Aryan:** Indo Aryan Mode- the beginning in Orissa- the Lingaraja at Bhubaneshwar. Hindu architecture at Raiputana(Temple of Surya, Osia, Marwar) and Gujrat (Temple of Surya, Modhera).The Khajuraho group-KhanariyaMahadev, Jain tample- Chaurmukh temple at Ranpur.	8
IV	**Later Dravidian Period:** Later Dravidian period- the vijayanagar and Madurai Dynasties – Noted temples at Hampi(Vitthala temple and HazaraRamam Temple). Madurai (Meenakshi temple) and Srinagar.	8
TOTAL		32

Text Books:

1. Brown, Percy. Indian Architecture (Buddhist and Hindu Period); D.B. Taraporevala Sons & Co. Pvt. Ltd., Mumbai.
2. Grover, Satish. Architecture of India: Buddhist and Hindu; Vikas Publishing House, New Delhi.
3. Fletcher, Banister. A History of Architecture; CBS Publishers & Distributors, New Delhi.
4. Tillotson, G.H.R. The Tradition of Indian Architecture: Continuity, Controversy and Change Since 1850; Yale University Press.

Reference Books:

1. Kramrisch, Stella. The Hindu Temple (Vol. 1 & 2); Motilal Banarsidass Publishers, Delhi.
2. Michell, George. The Hindu Temple: An Introduction to its Meaning and Forms; University of Chicago Press.
3. Hardy, Adam. The Temple Architecture of India; Wiley India Pvt. Ltd.
4. Rowland, Benjamin. The Art and Architecture of India: Buddhist, Hindu, Jain; Penguin Books.
5. Soundara Rajan, K.V. Art and Architecture of South India; Sundeep Prakashan, New Delhi.
6. Dhaky, M.A. The Indian Temple Forms in Karnata Inscriptions and Architecture; Abhinav Publications.

Reference Materials:

1. Journal of the Society of Architectural Historians (JSAH) – Relevant articles on Hindu temple development.
2. Marg Publications – Especially thematic issues on Indian temple towns and dynasties (e.g., Temples of India series).
3. Digital South Asia Library – Access to high-resolution temple plans and inscriptions.
4. ASI (Archaeological Survey of India) Reports – Documentation and research on protected monuments.

3rd Semester		
Paper 3	**Building Construction and Material III** **L-T-P-C: 1-0-4-5 Credits: 5** **Scheme of Evaluation: Studio**	**Subject Code: ARC132C312** **Level : 200**

Course Objective: The objective of **Building Construction and Material III (ARC132C312)** is to introduce students to the technicalities of building construction and material and to acquaint with various methods of construction.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define the types and construction methods of foundations including RCC and steel footings, and explain the design, planning, and treatment of timber and steel formwork systems.	BT1
CO 2	Explain masonry construction techniques including composite masonry and demonstrate the application of special concretes in varied construction contexts and joints.	BT2
CO 3	Apply knowledge of stair construction and materials and analyze the structural and architectural properties of steel used in modern building practices.	BT3
CO 4	Analyse the use of bamboo in construction by assessing its structural, aesthetic, and environmental qualities, including traditional and innovative joinery methods.	BT4

Prerequisite: NIL

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
IA	**FOUNDATIONS:** Types of foundations; isolated, combined, Raft, Base slab, Grillage, Introduction of Pile foundations. Principles and methods of construction of RCC & steel foundations and columns. **Formwork:** Design & planning of form work in timber, steel. Propping, centering, camber, cleaning, surface treatment	3	12
IB	**Material: Concrete:** Concrete Ingredients, grades of concrete, admixture, properties of concrete, production of concrete, mix proportioning, mixing, transporting, placing, compaction, curing of concrete and ready-mix concrete, sampling and testing of concrete, Construction joints, expansion joints, finish in concrete, chemical admixture. Uses of concrete for aesthetic & structural purpose.	1	
IIA	**MASONRY:** Construction & design of Masonry: Foundations with masonry. Composite Masonry, load bearing wall tiles, glass block masonry.	1	
IIB	**Material: Special Concrete:** Concreting under water, special concretes like light weight and high-density concrete. Construction joints, expansion joints, finish in concrete, chemical admixture. Uses of concrete for aesthetic & structural purpose	3	12
III A	**STAIRS:** Classification of stairs, Location and materials and technical terms.	1	
IIIB	**Material: Steel** Properties and architectural uses of mild steel and stainless steel for aesthetic & structural purpose	3	12

IV A	**Introduction to Bamboo** Types of Bamboo with joinery details	1	
IVB	**Material: Introduction to Bamboo** Properties and architectural aesthetic and structural uses and purposes, advantages, disadvantages	3	12
	TOTAL	64	

Text Books:

1. Building construction; Mackay, W.B. ;Building construction; 1st Ed.; 2005; Donheed; London.

Reference Books:

1. Chudley, R; Construction technology; 2nd Ed.; 1987; ELBS; Harlow.
2. Barry, R; Construction of building; 4th Ed.; 1999; East West Press; New Delhi.
3. Kumar, Sushil; Building Construction; 19th Ed.; 2001; Standard Publishers Distributors; Delhi.
4. Arora, Bindra; Building Materials

Reference Materials:

1. Journal of Construction and Building Materials Issue: Volume 279, 20 May 2021
2. Indian Concrete Journal Issue: May 2020
3. Journal of Steel Structures & Construction Issue: Volume 5, Issue 3, 2019
4. Journal of Bamboo and Rattan Issue: Volume 19, Issue 1, 2020

3rd Semester		
Paper 4	**Building Science I** **L-T-P-C: 2-0-0-2 Credits: 2** **Scheme of Evaluation: Theory**	**Subject Code:** ARC132C303 **Level : 200**

Course Objective: The objective of **Building Science I (ARC132C303)** is study the role of climate change in architectural design, the planning and construction of buildings with respect to climatic conditions and also varied materials impacting the human comfort level.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define the characteristics of climate change and familiarize student to design according to climatic conditions.	BT1
CO 2	Outline the basic elements and representations of climatic data.	BT2
CO 3	Discover the elements of thermal comfort and effects of landscape element on micro and macro site.	BT3
CO 4	Analyse on designing buildings as per various climatic zones of India	BT4

Prerequisite: NIL

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	**Introduction to the impacts of climate change:** Introduction to extreme events and gradual changes of the climate. To study human heat balance and comfort. To familiarize students with the design and settings for buildings for daylight and factors that influence temperature. Human comfort (body & thermal), thermal comfort factors and indices. Principles of thermal design, structural and ventilation controls and their application in building. Illumination and day lighting.Use of shading angles for lighting and radiation techniques and their application in buildings. To inform about the air pattern around buildings and the effect of wind on design of buildings and developing a site. Effect of climatic elements of thermal comfort environment.	12	0
II	**Elements of Climate:** Elements of climate, measurement and representations of climate data. Classification of tropical climates, major climate zones of India. Effects of landscape elements on site/micro climate.	12	0
III	**Thermal Comfort:** Body's heat exchange with surrounding environment. Thermal comfort indices viz., Effective temperature, bio-climatic chart etc., Kata thermometer and globe thermometer. Effects of landscape elements on site/micro climate.	12	0
IV	**Design Principles:** Design of buildings and developing a site. To expose the students to the various design strategies for building in different types of climatic zones in India. Exposure to different design strategies across the globe.	12	0
	TOTAL	48	

Text Books:

1. Manual of Tropical Housing and Building,Koenigsberger, O.H.; 1st Ed.; 2009; Orient BlackSwan; Hyderabad.
2. Climate design -Part I, Orient Longman, Madras

Reference Books:

1. Manakbhavan.,Bureau of Indian Standards IS 3792 (1987), Hand book on Functional requirements of buildings other than industrial buildings, (Part I – IV),.
2. Evans, Martin; Housing Climate and Comfort; 1st Ed.; 1980; Architectural Press, London.
3. Givoni, B; Man, Climate and Architecture; 1st Ed.; 1969; Applied Science Publishers Ltd., London.

Reference Materials:

1. Givoni, B; Passive and Low Energy Cooling of building; 1st Ed.; 1994; Van Nortrand Reinhold; New York.Gallo, C, Sala, M &Sayigh, A.M.M; Architecture, Comfort and Energy; 1st Ed.; 1988; Elsivier Science Ltd; New York.

3rd Semester		
Paper 5	**Building Services I** **L-T-P-C: 1-0-1-2 Credits: 2** **Scheme of Evaluation: Theory & Practical**	**Subject Code:** ARC132C317 **Level: 200**

Course Objective: The objective of **Building Services I (ARC132C317)** is to develop concepts of water management, sanitation and disposal to develop the designs with better services**.**

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Recall the basic concepts, sources, methods, and systems related to water supply, catchment, and treatment.	BT1
CO 2	Explain and interpret various water management systems including materials, fittings, pumps, and rainwater harvesting for urban and rural contexts.	BT2
CO 3	Apply principles of sanitation in designing drainage and pipe systems in buildings as per NBC and Indian Standards.	BT3
CO 4	Analyze waste, water and solid waste disposal methods, and evaluate treatment techniques for environmental sustainability.	BT4

Prerequisite: NIL

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	**Water Supply:** Introduction to water supply, Assessment of water requirements, Sources of supply, collection & method of supply. Catchment areas, reservoirs, recharge of ground water (methods) and their location. Roof top rainwater harvesting. Recovery of used water. Physical, Chemical and biological examination of water, Water treatment.	10	10
II	**Water Management:** Water pipes– materials (GI, PVC, CPVC/ UPVC pipes, introduction to copper plumbing, Pipe Accessories, Storage tanks, Pumps, Pipes – laying and jointing. Supply for a neighbourhood and town. Rainwater Harvesting, Recharging, Recycling and reuse, application in planning, water supply distribution systems (Urban & rural). Sheet work on water supply, overhead water tank and calculations.	10	10
III	**Sanitation: Sanitary pipes, fittings and fixtures, layout and design** Principles of sanitation, Study of Indian Standards and plumbing by-laws (NBC) Introduction to various sanitary pipes, joints, fittings and fixtures, their functions, placements and construction details. Study of internal and external drainage of various buildings. Single stack system, one pipe and two pipe systems, testing of house drains, Gradient used in laying drains and sewers, Self-cleaning and non-scoring velocities of drainpipes.	10	10
IV	**Sanitation: Wastewater treatment and disposal methods** Study of traps, inspection chambers, Manholes, Septic tanks, Soak pits and public sewage line. Study of disposal systems for domestic effluents from fitting to sewer line. Water waste-sewage disposal, primary treatment, secondary treatment and tertiary treatment. Modern types of sewage treatment plant **Solid waste, collections, treatment and disposal**	10	10

	Prevalent SWM practices and deficiencies, Storage of waste at source, collection, segregation and transportation of waste. Disposal of solid waste: Sanitary Land filling, Composting, Incineration, Pulverisation. Pyrolysis-advantages and disadvantages. Biogas systems and modern renewable energy systems. Sheet work on septic tank, soak pit, inspection chamber and calculations.		
	TOTAL	80	

Text Books:

1. Manual of Tropical Housing and Building,Koenigsberger, O.H.; 1st Ed.; 2009; Orient BlackSwan; Hyderabad.
2. Climate design -Part I, Orient Longman, Madras.

Reference Books:

4. Manakbhavan., Bureau of Indian Standards IS 3792 (1987), Hand book on Functional requirements of buildings other than industrial buildings, (Part I – IV).
5. Evans, Martin; Housing Climate and Comfort; 1st Ed.; 1980; Architectural Press, London.
6. Givoni, B; Man, Climate and Architecture; 1st Ed.; 1969; Applied Science Publishers Ltd., London.
7. Givoni, B; Passive and Low Energy Cooling of building; 1st Ed.; 1994; Van Nortrand Reinhold; New York.
8. Gallo, C, Sala, M &Sayigh, A.M.M; Architecture, Comfort and Energy; 1st Ed.; 1988; Elsivier Science Ltd; New York.

3rd Semester		
Paper 6	**Structures III** **L-T-P-C: 3-0-0-3 Credits: 3** **Scheme of Evaluation: Theory**	**Subject Code:** ARC132M306 **Level : 100**

Course Objective: The objective of **Structures III (ARC132M306)** is to enable students to understand advanced structural concepts involving bending, torsion, column stability, and structural stability in buildings and supporting elements. The course integrates theory with architectural design applications.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Recall and define fundamental concepts of bending stress and shear stress in various beam cross-sections.	BT1
CO 2	Explain and interpret the effects of combined direct and bending stresses, and the torsional behavior of shafts.	BT2
CO 3	Apply Euler's to analyze the strength and failure of columns and struts.	BT3
CO 4	Analyze the structural stability of buildings, dams, and retaining walls considering strength and stiffness.	BT4

Prerequisite:

Students must have completed Theory of Structures – II to understand fundamental stress-strain behavior, shear force, bending moments, and basic structural analysis.

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	**Bending Stress and Shear Stress:** Pure Bending Theory, Derivation & application of bending stresses in beams : Rectangular, circular, I & T beam section; Shear stresses & its distribution.	12	0
II	**Direct and Bending stress & Torsion:** Application of direct & bending stresses. Torsion of solid & hollow circular shafts of similar material; derivation & application.	12	0
III	**Columns and Struts:** Columns & struts; Euler's theory of hinged ends	12	0
IV	**Stability of Structures:** Concept of Strength and Stiffness of Structures and Structural Members. Stability of Buildings, Dams and Retaining walls; Stress derivation and application	12	0
	TOTAL	48	

Text Books:

1. Sanjay;Strength of materials, Bansal, R K and Bansal; 4th Ed.; 2009; Laxmi Publications, New Delhi.
2. S. Ramamrutham – Strength of Materials, Dhanpat Rai Publications, New Delhi
3. S.S. Bhavikatti – Basic Structural Analysis, Vikas Publishing House
4. Ramamrutham – Analysis of Structures I and II, Dhanpat Rai Publications

Reference Books:

1. Ramamrutham S.; *Strength of materials*; 16th Ed.; 2009; Dhanpat Rai; New Delhi
2. Hibbeler R.C. – Engineering Mechanics: Statics & Dynamics, Pearson Education
3. Vazirani & Ratwani – *Analysis, Design and Details of Structures*, 14th Ed., Khanna Publishers

3rd Semester		
Paper 8	**Architectural Presentation** **L-T-P-C: 1-0-1-2 Credits: 2** **Scheme of Evaluation: Practical**	**Subject Code:** **Level : 200**

Objective: The objective of Architectural Presentation (……) in techniques of three-dimensional presentation of built form through perspective drawing and rendering using different manual media. It aims to develop skills in visualizing interior and exterior spaces and presenting them professionally.

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Recall principles of perspective drawing and spatial depth representation	BT1
CO 2	Understand one-point, two-point and multi-point perspective construction	BT2
CO 3	Apply perspective techniques to interior and exterior built forms	BT3
CO 4	Develop fully rendered presentation drawings using various media.	BT4

Prerequisite: NIL

Detailed Syllabus:

Modules	*Topics (if applicable) & Course Contents*	*Periods/ Hours*	
		L	***P***
I	**Principles of Perspective & One Point Perspective** • Introduction to three-dimensional visualization • Horizon line, eye level, station point, vanishing point • One-point perspective construction • Interior perspectives: ○ Hotel Lounge ○ Restaurant interiors ○ Corridor views • Depth, scale, proportion and furniture detailing	0	16
II	**Two Point & Multi Point Perspective** • Two-point perspective construction (angular views) • Exterior building perspectives • Built-form with landscaping elements • Multi-point perspective (complex built forms, shopping mall interiors, atriums) • Human figures, vehicles, vegetation in perspective	0	16
III	**Rendering Techniques: Pencil & Charcoal** • Tonal values and shading techniques • Light and shadow study • Texture rendering (glass, brick, concrete, vegetation, water etc.) • Perspective rendering using: ○ Pencil (HB to 8B techniques) ○ Charcoal (smudging, highlights, depth enhancement) • Composition and presentation layout	0	16
IV	**Rendering Techniques: Pen & Ink / Water Colour** • Line weight variation • Hatching and cross-hatching • Wash techniques	0	16

	• Water colour techniques: ○ Wet-on-wet ○ Dry brush ○ Layering and gradients • Final presentation plates • Portfolio compilation		
	TOTAL	64	

Text Books:

1. Rendering with pen and ink.

3rd Semester		
Paper 8	**Computer Application I** **L-T-P-C: 0-0-2-2 Credits: 2** **Scheme of Evaluation: Practical**	**Subject Code:** ARC132S318 **Level : 200**

Objective: The objective of **Computer Application I (ARC132S318)** is to familiarize students with drawing aids and equipment along with learning to computerise design drawings and orient students to the digital tools.

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Relate to the applications of computers in architecture .	BT1
CO 2	Discuss the usage of applications of drawing and modification commands.	BT2
CO 3	Interpret how to use software commands.	BT3
CO 4	Analyze the methods of digital representation.	BT4

Prerequisite: NIL

Detailed Syllabus:

Modules	*Topics (if applicable) & Course Contents*	*Periods/ Hours*	
		L	***P***
I	**Interface:** Introduction to the interface, Digital drawings tools, drawing lines and shapes, modifying lines and shapes, drawing with accuracy and speed tools and commands. Page setup.	0	16
II	**Drawing and Modifying:** Drawing and Modifying objects, properties, Units, dimensions, lines and pen weight. Blocks, array.	0	16
III	**Introduction to Commands:** Basic drawing commands, editing commands, scaling, setting dimensioning variables etc.	0	16
IV	**Presentation and plotting:** Presentation, hatching and rendering, texts, Layers, planes , views and viewports Import/export, layout , xref, print setup and plotting(with a specific project submission)	0	16
	TOTAL	64	

Text Books:

2. Up and Running with AutoCAD 2015: 2D & 3D Drawing and Modelling,Gindis, E;1st Ed.; 2014; Elsevier; London.

Reference Books:

1. Seidler, D. R.; Digital Drawing for Designers: A Visual Guide to AutoCAD 2012; 1st Ed.; 2007; Fairchild Publications; London.
2. Smith, B. L.; 3ds Max 2008 Architectural Visualization Beginner to Intermediate; 1st ed.; 2007; 3DATS; China.

Reference Materials:

1. Moss, Elise; Autodesk AutoCAD Architecture 2016 Fundamentals; 1st Ed.; 2011; SDC Publications; London.
2. Omura, George & Benton, Brian C; Mastering AutoCAD 2016 and AutoCAD LT 2016; 1st Ed.; 2017; John Wiley & Sons; London.

4th Semester		
Paper 1	**Architectural Design IV** **L-T-P-C: 1-0-7-8 Credits: 8** **Scheme of Evaluation: Studio**	**Subject Code:** ARC132C411 **Level : 100**

Course Objective: The objective of **Architectural Design IV(ARC132C411)** is to orient students towards contemporary architecture with special emphasis on sustainability.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand the principles of sustainable development and analyze architectural case studies using sustainability parameters like LEED or IGBC	BT1
CO 2	Apply sustainability criteria and regulatory norms to develop conceptual design and site planning solutions for architectural projects.	BT2
CO 3	*Design detailed architectural plans incorporating building services, materials, and construction techniques with a focus on innovation and sustainability.*	BT3
CO 4	Prepare and present professional-level architectural drawings and portfolios demonstrating the execution-ready design process.	BT4

Prerequisite: nil

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	**Sustainable development and case study:** A detailed study of sustainability with case studies. Site to be identified and mapped in terms of LEED or IGBC	1	31
II	**Concept development and site planning:** Concept development of the project and detailed site planning considering sustainability criteria and site planning considering local planning and development norms and building byelaws	1	31
III	**Detailed planning:** Detailed building planning with services and construction techniques considered	1	31
IV	**Representation drawings:** Finalization of design and make presentation drawing for execution. Presentation of works of design through portfolio or panels	1	31
	TOTAL	128	

Remarks:

Design of Vocational training centre, resort, school etc.

- The design problem is to be selected in such a way that it lets the student explore several themes on contemporary architecture. The design should also encourage the use of both locally available materials with new innovation techniques and new building material in a sustainable manner. Projects such as eco-resorts, civic buildings, institutions etc. could be taken up.

Text Books:

1. Time saver standards for building types,De Chiara, Joseph and Crosbie, Michael J; 2nd Ed.; 2011; Tata McGraw Hill; New Delhi.

Reference Books:

1. Bhatia, Gautam; Lauri Baker: Life, Works and Writings; 1st Ed.; 1994, Penguin Books India Limited; New Delhi

4th Semester		
Paper 2	**History of Architecture III** **L-T-P-C: 2-0-0-2 Credits: 2** **Scheme of Evaluation: Theory**	**Subject Code:** ARC132C404 **Level: 200**

Course Objective: The objective of **History of Architecture III (ARC132C404)** is to provide an understanding of the evolution of Islamic and Colonial Architecture in India in their various stylistic modes characterized by technology, ornamentation, and planning practices.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define the key architectural features, construction techniques, and stylistic elements of the Imperial Style in Islamic architecture as developed under the Slave, Khilji, Tughlaq, Sayyid, and Lodi dynasties through detailed study of structures like Qutub Minar, Alai Darwaza, and tombs of rulers.	BT1
CO 2	Outline the distinguishing characteristics of Provincial Style architecture by identifying regional adaptations, material use, ornamentation, and spatial arrangements in monuments of Ahmedabad, Jaunpur, Bengal, and Bijapur, such as Jami Masjid, Gol Gumbaz, and Eklakhi Tomb.	BT2
CO 3	Utilize an understanding of Mughal architectural principles—symmetry, scale, axial planning, and decorative systems—to study prominent structures built during the reigns of Humayun, Akbar, Jahangir, Shah Jahan, and Aurangzeb, including Humayun's Tomb, Taj Mahal, and Fatehpur Sikri.	BT3
CO 4	Analyse the transformation in Indian architectural landscape brought by colonial influence by examining early colonial buildings of the East India Company era and evaluating the urban planning and architectural contributions of Edwin Lutyens and Herbert Baker to New Delhi.	BT4

Prerequisite: NIL

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	**Imperial Style:** Slave, Khilji, Tughlaq, Sayyid& Lodi dynasties. Eg.Quwaat-Ul-Islam Mosque, Qutb-Minar, Enlargement of Quwaat-Ul-Islam Mosque by Iltutmish, Tomb of Iltutmish, Enlargement of Qutub complex by Ala-Ud-din Khilji, Alai Darwaza, Tomb of Ghaias-Ud-in Tughlaq, Khirki Masjid, Delhi, Tomb of Feroz Shah Tughlaq, Shish Gumbad& Tomb of Mubarak Shah Sayyid.	8	0
II	**Provincial Style:** Ahmedabad, Jaunpur, Bengal and Bijapur Eg.Atala masjid Jaunpur, Jami Masjid, Jaunpur, Jami Masjid, Biapur, Ibrahim Rauza, Bijapur. GolGumbaz, Bijapur, Adin Masjid, Pandua, Eklakhi tomb, Pandua, DakhilDarwaza, Gaur, Jami Masjid Ahmedabad	8	0
III	**Mughal Period:** Monumental buildings in the regime of Humayun, Akbar, Jehangir, Shahjahan&Aurangazed. Eg.Humayan's tomb, FatehpurSikri (layout, Jami Masjid, BulandDarawaza Tomb of SalimChistiDiwaneKhas) Akbar's	8	0

	Mausoleum TajMahal.		
IV	**Arrival of British:** Early colonial period,monumental buildings executed in the regime of East India Company up to middle of 19th century. Eg.St. Paul's Cathedral, Calcutta & Bombay Town Hall. Later colonial period – Contribution of Edwin Lutyens and Herbert Baker to the layout and architecture of New Delhi.	8	0
	TOTAL	32	

Textbooks:

1. Time saver standards for building ty "History of Architecture in India" by Tadgel Christopher.

Reference Books:

1. Brown, Percy "Indian Architecture – Islamic period"

Reference Materials:

1. Journal of the Society of Architectural Historians (JSAH)
2. Journal of Interior Design, Vol. 47, Issue

4th Semester		
Paper 3	**Landscape Architecture** **L-T-P-C: 2-0-1-3 Credits: 3** **Scheme of Evaluation: Theory + Practical**	**Subject Code:** ARC132D413 **Level : 200**

Course Objective: The objective of **Landscape Architecture (ARC132D413)** is to introduce students to the discipline of landscape architecture and its relevance to Architecture, to gain an insight into the changing relationship of human with nature. This course shall have a direct application in the design studio of the same semester as well as subsequent semesters for site planning and landscape design of the respective design assignments.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand the fundamental concepts of landscape and landscape architecture and analyze the evolving relationship between humans and nature in varied landscape contexts.	BT1
CO 2	Interpret historical developments in landscape architecture and evaluate notable projects and styles across cultures and time periods.	BT2
CO 3	Identify and apply natural and man-made elements of landscape architecture in basic design exercises, incorporating principles of landscape design and field documentation.	BT3
CO 4	Demonstrate site analysis, planning techniques, and integrate landscape design within site development projects at neighborhood or architectural scale.	BT4

Prerequisite:

- Basic Concept of Art

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	**Introduction to Landscape & Landscape Architecture** • Landscape – its meaning, Experience of a landscape, Aesthetics & Imagery of a landscape, Relationships of humans and nature; how the Scales and Conception of landscapes evolve over time, Sense of place in a Landscape. • Landscape Architecture – definition, related terminologies and Scope.	12	4
II	**Elements of Landscape Architecture & their application in Design** • Natural elements of landscape architecture *(landforms, water, vegetation)*, Natural systems. • Man-made elements, modifications in natural systems with man-made elements, co-existence of natural and man-made elements. • Principles of Design incorporating these landscape elements. • Documentation – *Field identification of common Indian trees, plants, shrubs etc.*	12	4
III	**Historical Overview of Site Planning and Landscape Architecture** • Study of Historical Landscape Gardens – *Chinese & Japanese, Persian & Mughal, English, Italian, French and Renaissance gardens.* • Study of 19th & 20th Century noted Landscape Arch. Projects *(Garden of Versailles, Stowe, Central Park, Copacabana Promenade, Sunder Nursery etc.)* by renowned Landscape	12	4

	Architects – *Lancelot Brown, André Le Nôtre, Frederick Law Olmstead, Geoffrey Jellicoe, Gertrude Jekyll, Roberto Burle Marx, Martha Schwartz, Dan Kiley, Mohammad Shaheer, Kishore Pradhan, Prabhakar Bhagawat, Lawrence Halprin, Michael Van Valkenburg etc.*		
IV	**Site Analysis & Site Planning** • Site Analysis – definition, its need and stages/ layers, Site Inventory – *Topography, Vegetation, Soil, Hydrology, Climate etc.* • Principles of Site Planning, Design Issues in planning process, Siting of buildings, integrating the built and open spaces etc. • Comprehensive Site development, materials, street, site furniture etc. • Design Studio on Site Plan with Landscape Design (Neighbourhood Scale) *OR* Studio component of the Semester may be integrated with Architectural Design of the current semester.	12	4
	TOTAL	64	

Text Books:

1. Landscape Architecture: Simmonds J.O.; 5 edition (1 March 2013), McGraw-Hill Education.
2. The Experience of Landscape: Appleton (1996), Wiley.
3. Introduction to Landscape Architecture: Laurie Michael, 2nd Revised edition (1 February 1986), Elsevier Science Ltd.
4. Landscape Architecture in India – A Reader: Mohammad Shaheer (1 January 2013), LA, Journal of Landscape Architecture.
5. Form and Fabric in Landscape Architecture – A Visual Introduction: Catherine Dee (2001), Spon Press.
6. Site Planning: Kevin Lynch, Gary Hack (1962), MIT Press.

Reference Books:

1. Krishen Pradip; Jungle Trees of Central India; Penguin Random House India Pvt. Ltd, 2013.
2. Krishen Pradip; Trees of Delhi; Penguin Books India Pvt Ltd, 2006.
3. Grant W. Reid; Landscape Graphics; Whitney Library of Design, Watson Guptill Publications, New York, 2002. Charles W Harris and Nicholas T. Dine; Time Saver Standards for Landscape Architecture; 2nd Edition, Mcgraw – Hill, International Edition, Arch. Series, 1988.

4th Semester		
Paper 4	**Building Construction and Material IV** **L-T-P-C: 1-0-4-5 Credits: 4** **Scheme of Evaluation: Studio**	**Subject Code:** ARC132C412 **Level : 200**

Course Objective: The objective of **Building Construction and Material IV (ARC132C412)** is to introduce students to the technicalities of building construction and material and to acquaint them with various methods of construction.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define the construction techniques and reinforcement systems for various RCC roofs and retaining walls, including the application of paints and varnishes on surfaces.	BT1
CO 2	Describe the purpose and detailing of RCC components such as beams and retaining walls, and explain the role of polishes, grouts, and bonding agents in concrete finishes.	BT2
CO 3	Demonstrate the use of precast and prefab construction systems, ferrocement, and filler slabs as alternatives to RCC, with appropriate waterproofing techniques.	BT3
CO 4	Analyse the structural use and joinery details of bamboo and assess the suitability of flooring materials, tiles, adhesives, and finishes in building construction.	BT4

Prerequisite: NIL

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
IA	**RCC ROOFS:** Study of principles and methods of construction of RCC one-way, two-way slabs, ribbed slab, filler slab, flat slab, sloping RCC roof, vaults, and domes, including form-work techniques and reinforcement details with understanding the functions of RCC beams.	2	24
IB	**MATERIAL: Paints:** Paints, varnishes and distempers, emulsions, cement base paints. Constituents of oil paints, characteristics of good paints, types of paints and process of painting different surfaces.	1	1
IIA	**RCC RETAINING WALL:** Designing and stability of RCC retaining and breast wall.	1	7
IIB	**MATERIAL: Polishes:** Types of varnish, methods of applying varnish and fresh polish and melamine finish. Grouts and anchors, repairs and protective coatings, bonding agents	1	1
III A	**ALTERNATIVES TO RCC:** Precast and Prefab components Masonry blocks, hollow blocks, jallis, shelving units, hollow clay blocks roofing techniques (filler slab), Expansion joints – Necessity, location, and detailing Ferro cement elements, Bamboo Joineries & Construction details, polishes, finishes	2	14
IIIB	**MATERIAL: Water proofing:** sealants and water proofing and weather proofing compounds, weather proofing compounds	1	0
IV A	**ALTERNATIVES TO RCC:** Bamboo Bamboo Joineries & Construction details, polishes, finishes, Bamboo as a structural material for columns	1	7

IVB	**MATERIAL: flooring types & finishes:** Flooring types, Paving, and interlocking tiles, adhesives for fixing floor finishes, and joint filers	1	0
	TOTAL	64	

Text Books:

1. Building construction; Mackay, W.B.; Building construction; 1st Ed.; 2005; Donheed; London.
2. Building Construction; Kumar, Sushil; 19th Ed.; 2001; Standard Publishers Distributors; Delhi.

Reference Books:

1. Chudley, R; *Construction technology*; 2nd Ed.; 1987; ELBS; Harlow.
2. Barry, R; *Construction of building*; 4th Ed.; 1999; East West Press; New Delhi.

Reference Materials:

1. Construction and Building Materials, Volume 279, May 2021 – Elsevier
2. Indian Concrete Journal, May 2020 – Topic: RCC detailing, admixtures & bonding agents.
3. Journal of Steel Structures & Construction, Volume 5, Issue 3, 2019
4. Journal of Bamboo and Rattan, Volume 19, Issue 1, 2020 – Topic: Bamboo joinery systems and finishes

4th Semester		
Paper 5	**Structures IV** **L-T-P-C: 3-0-0-3 Credits: 3** **Scheme of Evaluation: Theory**	**Subject Code:** ARC132M405 **Level : 100**

Course Objective: The objective of **Structures IV (ARC132M405)** is to introduce students to structural concepts involving beam deflection and the fundamentals of cement, aggregates, concrete workability, and reinforced cement concrete (RCC) design.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Recall basic concepts of beam deflection, moment distribution, fixed beams, and three-hinged arches based on IS 875 standards.	BT1
CO 2	Understand types, grades, and properties of cement and aggregates, and their impact on concrete quality and strength.	BT2
CO 3	Apply knowledge of concrete properties such as batching, mixing, curing, workability, and strength through standard procedures and field practices.	BT3
CO 4	Analyze basic RCC design concepts under Limit State Design, and differentiate between working stress, ultimate load methods, and design parameters from IS 456:2000.	BT4

Prerequisite:

*Students must have completed **Theory of Structures – III** to understand bending, torsion, column behavior, and basic stability before progressing to reinforced structures and concrete behavior.*

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I (Theory)	**Beam Deflection:** Application of Deflection of simply Supported and cantilever beams by double integration Method. Application of Analysis of fixed beams with u. d. 1. &. Concentrated loads.	9	0
II (Theory)	**Cement and Aggregates:** Types & grades of Cements, testing of cements Introduction to IS 456-2000. Aggregates; Coarse & fine, characteristics & effect on its strength; particle shape& texture, size & grading; Moisture contents in aggregates, bulking of sand. Requirement of water for concrete	9	0
III (Theory)	**Concrete:** Batching & mixing of concrete, workability- slump of concrete w/c ratio, voids & permeability. Conveyance of Concrete, placing & compacting, curing, compressive strength of concrete, testing of concrete. Relation of durability & cover of concrete.(repeated in building construction)	9	0
IV (Theory)	**Introduction to RCC Design:** Introduction to Limit State Design, Introduction to IS 456:2000. Brief understanding of Working Stress Method, Ultimate Load Method Concepts of Characteristic Load, Characteristic Strength, Factors of Safety, Load Combinations, Stress-Strain relationships of materials for design consideration etc.	9	0
	TOTAL	36	

Text Books:

1. *Sanjay;Strength of materials, Bansal, R K and Bansal; 4th Ed.; 2009; Laxmi Publications, New Delhi.*
2. *S. Ramamrutham – Strength of Materials, Dhanpat Rai Publications, New Delhi*
3. *S.S. Bhavikatti – Basic Structural Analysis, Vikas Publishing House*
4. *Ramamrutham – Analysis of Structures I and II, Dhanpat Rai Publications*

Reference Books:

1. Ramamrutham S.; *Strength of materials*; 16th Ed.; 2009; Dhanpat Rai; New Delhi
2. Hibbeler R.C. – Engineering Mechanics: Statics & Dynamics, Pearson Education
3. Vazirani & Ratwani – *Analysis, Design and Details of Structures*, 14th Ed., Khanna Publishers

4th Semester		
Paper 6	**Building Services II** **L-T-P-C: 2-0-1-3 Credits: 3** **Scheme of Evaluation: Theory + Practical**	**Subject Code:** ARC132M416 **Level : 200**

Course Objective: The objective of **Building Services II (ARC132M416**) is to develop concepts of design, installation, operation and monitoring of the electrical services in buildings.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Recall the principles and functions of mechanical, electrical, and plumbing systems in buildings.	BT1
CO 2	Explain various types of building services such as light and its source.	BT2
CO 3	Apply design guidelines for integration of electrical system and fittings in architectural layouts	BT3
CO 4	Analyze electrical layout in buildings and optimize planning strategies for specific building typologies.	BT4

Prerequisite: Nil

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	**Light:** Principles of light – Electromagnetic radiation, waves, nature of vision, measurement of lighting. Principles of illumination: definitions, Visual tasks, Factors affecting visual tasks Units of light, definitions of flux, solid angle, luminous intensity – utilization factor – depreciation factor- MSCP – MHCP, brightness, glare. Electric light sources: brief description, characteristics and application of different types of lamps, methods of mounting and lighting control Luminaries classification/ - Lumen method for design – Room reflectance/ Glare-manufacturer's data on luminaries/luminaries' cost	17	5
II	**Lighting in buildings:** Light and its sources, lighting criteria, the visual field, daylight prediction methods Artificial lighting, lighting levels for various activities, Calculation for lighting levels.	17	5
III	**Electrical system and fittings:** Laws of electrical circuits: Ohms and Kirchhoff's Laws, Basics of electricity – Single/Three phase supply. Earthing for safety – types of earthing – ISI specifications, Electrical wiring systems in domestic and commercial buildings. Conduits, Types of wiring Diagram for connection. Bus way, Bus Bars, lighting track and conduits (Aluminium metallic, non-metallic) arrangements. Power handling, equipment, switch board, distribution board, mains, fuse, meter, circuit breaker etc. panel boards. Lighting conductors: Electronic and Communication systems, Electrical Installations in Buildings.	17	5
IV	**Electrical layout in buildings:** Main and distribution boards – transformers – switch gears – substations – space requirement and Layout of the same in building types. Planning and layout of installations within a building complex. Different meters and protection units. Different type of loads and their individual protections. Selection	19	5

	of cable/wire sizes; potential sources of fire hazards and precautions. Emergency supply-standby (generators, invertors) & UPS. A specific design problem on this aspect.		
	TOTAL	90	

Text Book:

1. Ambrose, E.P.; *Electric Heating;* 1st Ed,; 1968; John Wiley& Sons; New York.
2. Philips, Derek; *Lighting in Architectural Design;* 1st Ed.; 1964; McGraw Hill; New York.
3. Hopkenson, R.G. & Kay, J.D.; *The lighting of Buildings;* 1st Ed.; 1969; Faber & Faber; London

Reference Books:

1. Elevators, Escalators, Moving Walkways – Manufactures catalogues
2. *National Building Code.*
3. *Electrical systems.*
4. Handbook of building Engineers in metric systems; 1968; New Delhi
5. *National Building Code.*

4th Semester		
Paper 8	**Computer Application II** **L-T-P-C: 0-0-2-2 Credits: 2** **Scheme of Evaluation: Practical**	**Subject Code:** ARC132S417 **Level : 200**

Course Objective: The objective of **Computer Application II (ARC132S417)** is to familiarize students with drawing aids and equipment along with learning to computerise design drawings, to orient students to the digital tools.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**BT Level**
CO1	Identify various tools, plugins and extensions in software such as AutoCAD, Photoshop and SketchUp used in architectural representation.	BT1
CO2	Understand the significance of computerised design tools in the graphical and digital representation of architectural objects and spaces.	BT2
CO3	Use AutoCAD, Photoshop, and SketchUp to create accurate architectural drawings, 3D models and rendered views.	BT3
CO4	Analyse and compare different digital tools and techniques to determine the most effective method for solving architectural modelling problems and representing design details.	BT4

Detailed Syllabus:

Modules	**Course content**	**Periods**	
		L	**P**
I	**AutoCAD Revision and Advance:** Introduction to the interface, Digital drawings tools, drawing lines and shapes, modifying lines and shapes, drawing with accuracy and speed tools and commands. Page setup. Drawing and Modifying objects, properties, Units, dimensions, lines and pen weight. Blocks, array. Presentation, hatching and rendering, texts, Layers, planes , views and viewports Import/export, layout , xref, print setup and plotting (with a specific project submission)	2	14
II	**Introduction to photoshop (advance):** Photoshop Digital Rendering Techniques	2	14
III	**Introduction to SketchUp/Revit;** Basic tools, Visualisation, shortcuts of SketchUp	2	14
IV	**Problem Solving with detailed modelling** Creating Plans, Elevation, Section and 3d models using AutoCAD, Photoshop, SketchUp / Revit	2	14
Total		64	

Text Books:

1. *Up and Running with AutoCAD 2015: 2D & 3D Drawing and Modelling*,Gindis, E;*1st Ed.; 2014;* Elsevier; London.
2. *Digital Drawing for Designers: A Visual Guide to AutoCAD 2012;* Seidler, D. R.; 1st Ed.; 2007; Fairchild Publications; London.

Reference Books:

1. Smith, B. L.; *3ds Max 2008 Architectural Visualization Beginner to Intermediate;* 1st ed.; 2007; 3DATS; China.

2. Moss, Elise; *Autodesk AutoCAD Architecture 2016 Fundamentals;* 1st Ed.; 2011; SDC Publications; London.
3. Omura, George & Benton, Brian C; *Mastering AutoCAD 2016 and AutoCAD LT 2016;* 1st Ed.; 2017; John Wiley & Sons; London.

5th Semester		
Paper 1	**Architectural Design V** **L-T-P-C: 1-0-7-8 Credits: 8** **Scheme of Evaluation: Studio/Jury**	**Subject Code:** ARC132C511 **Level : 300**

Course Objective: The objective of **Architectural Design V (ARC132C511)** is to understand the need for creating architecture as an envelope to system dependent program, to identify and understand the role of services in the design of buildings; significance of material and construction techniques; climatic factors, to Introduce students to development regulations (building byelaws and rules); circulation networks (people, vehicular access), site planning etc.

Course Outcome:

CO Level	Course Outcome	BT Level
CO1	Outline key principles and strategies of sustainable and contemporary architecture relevant to group housing design.	**BT1**
CO2	Understand how sustainable design principles can be integrated into architectural projects through the use of local and innovative building materials.	**BT2**
CO3	Apply knowledge gained from literature reviews and case studies to develop context-sensitive concepts, architectural solutions and site planning strategies for group housing projects.	**BT3**
CO4	Analyse alternative architectural solutions, site conditions, and material choices to formulate detailed design solutions and representation drawings that reflect sustainable and contemporary architectural thinking.	**BT4**

Detailed Syllabus:

Modules	Course content	Periods	
		L	**P**
I	**Literature review and case studies:** Learning from detailed study and analysis of building systems and envelopes; character of group housing through literature review and visiting buildings in varied settings.	2	30
II	**Concept development and site planning:** Concept development of the project and detailed site planning considering sustainability criteria and site planning considering local planning and development norms and building byelaws.	2	30
III	**Detailed planning:** Detailed building planning with services and construction techniques considered. The Design Studio will give prominence to bridging the gap between innovations in materials and techniques of construction.	2	30
IV	**Representation drawings:** Finalization of design and make presentation drawing for execution which includes Site plan, floor plans, elevation, sections, details,3D Perspectives and models.. Presentation of works of design through portfolio or panels.	2	30
TOTAL		128	

Remarks:
Design of Group housing project (HIG, MIG, LIG including multistoried apartments), etc

Text Books:

1. *Time saver standards for building types*,De Chiara, Joseph and Crosbie, Michael J*;* 2nd Ed.; 2011; Tata McGraw Hill; New Delhi.

Reference Books:

1. Herg Cross Rudolph *Neuferts Architects Data*, Lockwood and Sons
2. National Building code (Latest version

5th Semester		
Paper 2	**History of Architecture IV** **L-T-P-C: 2-0-0-2 Credits: 2** **Scheme of Evaluation: Theory**	**Subject Code:** ARC132C503 **Level : 300**

Course Objective: The objective of **History of Architecture IV (ARC132C503)** is to provide an understanding of Western Architecture during Renaissance, Baroque, Neo Classical and Modern Periods.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Recall the background, features, and key examples of Renaissance and Baroque architecture.	BT1
CO 2	Understand the transitional trends leading to modernism, including Palladian, Greek, and Gothic Revival styles	BT2
CO 3	Apply knowledge to examine the impact of the Industrial Revolution on architecture, society, and technology.	BT3
CO 4	Analyze the principles and contributions of major architectural movements and masters of the Modern era.	BT4

Prerequisite: NIL

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	**Renaissance Period:** Background and influences on Renaissance Architecture, Characteristics of Renaissance Architecture in general. St. Andrea, Mantua , Palazzo Rucellai by Leon alberti, Palazzo Rucellai by Leon alberti, Villa Rotunda (Capra) by Palladio, St. Peter's Rome by Michelangelo and others, St. Paul's London by Sir Christopher Wren. **General characteristics of Baroque.** St.Peter's Plazza by Berinini.	8	0
II	**Transitional Period:** A brief account of the situation before the changeover to Modern architecture in Europe, Palladian revival in Britain, Greek Revival and Gothic Revival. e.g. Chiswick House, London (Palladian Style - William Kent), Mereworth castle, Kent, England (Palladian Style - Colen Campbell) etc.	6	0
III	**Impact of Industrial Revolution in Europe:** The Social, economic and political changes effected, New requirements of the society, New materials and technology developments.	8	0
IV	**a) Modern era:** The Chicago School - works of Louis Sullivan & Early Industrial buildings, Contributions of Bauhaus, De Stiji movement, Italian Futurism, Art Nouveau movement, Prudential/Guaranty Building., Buffalo, NY (Louis Sullivan and Danmar Adler) **b) Works of Great Masters:** Le Corbuster, FL Wright, Ludwig Mies Van Der Rohe, Walter Gropius, Oscar Niemeryer e.g. Villa Savoye, France, Falling Waters, Pennsylvania, Seagram Building, Park Avenue, New York, Fagus Factory, Germany, Alvorada Palace, Brazil.	10	0

	TOTAL	32

Text Books:

1. History of Architecture in India" by Tadgel Christopher.

Reference Books:

1. "Modern Architecture – A Critical History" by Frampton Kenneth
2. "A history of Architecture" by Fletcher, Bannister

5th Semester		
Paper 3	**Theory of Architecture** **L-T-P/S-C –2-0-0-2 Credits: 2** **Scheme of Evaluation: Theory**	**Subject Code:** ARC132C508 **Level : 100**

Course Objective: The objective of **Theory of Art and Architecture (ARC 132C508)** is to study the role of culture and art on architecture in Indian context, to study of visual art principle, monumental and human scale, to study of Impact of religious philosophy on the physical form

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**BT Level**
CO 1	Define the evolution of architecture in India through its various style of Architecture.	**BT1**
CO 2	Outline the Cultural and Religious implications on Architecture.	**BT2**
CO 3	Develop an understanding of Basic Principles of Architectural Composition, Different Types of Spatial Organization, Theory in Antiquity and Renaissance; 18th Century; 19th Century, Modern Movement, Post Modern Theory,	**BT3**
CO 4	Analyze the Contribution to New Architectural Thinking, Positive and Normative Theory.	**BT4**

Prerequisite:

- No prerequisite required

Detailed Syllabus:

Modules	*Topics (if applicable) & Course Contents*	*Periods/ Hours*	
		L	***P***
I	**(a) Basic Principles of Architectural Composition;** **(b) Different Types of Spatial Organization:** (a) Proportion, Scale, Symmetry, Hierarchy, Rhythm, Contrast, Harmony (b) Linear, Centralized, Radial, Clustered and Grid	**8**	**0**
II	**Theory in Antiquity and Renaissance; 18th Century; 19th Century:** Vitruvius, Alberti; Laugier, Boullee, Ledoux; Violet Le Duc, John Ruskin, Gottefried Semper	**8**	**0**
III	**(a) Modern Movement; (b) Post Modern Theory** **(c) Contribution to New Architectural Thinking**: (a) Ideas of Adolf Loos and EeroSarinen; (b) Robert Venturi and Charles Jencks (Library Study) (c) Christopher Alexander	**8**	**0**

IV	**Positive and Normative Theory:** Privacy Theory, Design Process, Proxemics Theory, Anthropometry, Ergonomics	**8**	**0**
	TOTAL	32	

Text Books:

1. "Form Space and Order" by Francis D.K Chin.
2. "A Catalogue of Modern Books on Architecture, Theoretical, Practical, and Ornamental" by J. Taylor.

Reference Books:

1. Taschen; Theory of Architecture
2. Jon Lang; Positive and Normative Theory

5th Semester		
Paper 4	**Building Construction and Material V** **L-T-P-C: 1-0-4-5 Credits: 4** **Scheme of Evaluation: Studio**	**Subject Code:** ARC132C512 **Level : 200**

Course Objective: The objective of **Building Construction and Material V (ARC132C512)** is to introduce students to the technicalities of building construction and material and acquaintance with various methods of construction.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define the basic working of Pile Foundation and Anti – Termite Treatment.	BT1
CO 2	Classify the different types of Beams and Glass Materials	BT2
CO 3	Discover the fixing and construction details of Aluminium Glass Doors and windows along with application of glass finishes.	BT3
CO 4	Analyze various aspects of Panelling and glass fixing technique in the practical field.	BT4

Prerequisite: Nil

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I A	PILE FOUNDATION: Types of Pile foundations; Cast-in situ Piles, Precast Piles, Sheet, Friction Piles, Types of Pile caps showing detailed Reinforcement fixing & method of driving Piles.	15	5
IB	Material: ANTI-TERMITE: Anti-termites Ingredients, Properties of Anti-termites, mixing proportioning and its properties.	10	30
II A	RCC BEAMS: Understanding functions of beams and their importance, Types of beams- Simply supported, Cantilever &Hunched beam showing with reinforcement detail & formwork techniques for real world application.	10	30
II B	Material: GLASS Manufacturing process of Glass, Various classification of Glasses- Clear glass, Tinted Glass, Frosted Glass, Fibre Glass & Glass blocks, Installation / fixing details.	15	5
III A	ALUMINIUM DOORS & WINDOWS: Aluminium doors with normal hinge fixing, with floor spring, Frameless door, Aluminium Casement Window, Aluminium Sliding window (two & three tracks with understanding of Anodised & Powder coated section).		
III B	Material: GLASS FINISHES: Understanding the technique of glass finishes- Polishing & Bevelling. Coloured Glass- Itching of Glass, its types and process of Itching & other finishes-Frosted film & fixing process.		
IVA	PARTITIONS& PANELLING: Understanding the types of Partitions- Solid Partitions, Glazed Partition & Low height Partition showing fixing details with aluminium section. Panelling with different materials- Ply, Gypsum, Bison Panel & Acoustical element fixing with aluminium sections.		

IVB	Material: GLASS FIXING: Knowledge of fixing techniques of glass- Patch fitting with various types of Patch fittings , Different types of Brackets & d-clips, Spider fittings of various legs and joint filler of glasses with silicon.		
	TOTAL	120	

Text Books:

1. Building construction; Mackay, W.B.; 1st Ed.; 2005; Donheed; London.
2. Building Construction; Kumar, Sushil; 19th Ed.; 2001; Standard Publishers Distributors; Delhi.

Reference Books:

1. Chudley, R; Construction technology; 2nd Ed.; 1987; ELBS; Harlow.
2. Barry, R; Construction of building; 4th Ed.; 1999; East West Press; New Delhi.

Reference Materials:

1. Kumar, Sushil; Building Construction; 19th Ed.; 2001; Standard Publishers Distributors; Delhi.
2. Arora, Bindra; Building Materials

5th Semester		
Paper 5	**Building Services III** **L-T-P-C: 1-0-2-3 Credits: 4** **Scheme of Evaluation: Theory + Practical**	**Subject Code:** ARC132C514 **Level : 300**

Course Objective: The objective of **Building Services III (ARC132C514**) is to develop an understanding of advanced building services such as air conditioning, elevators, escalators, and fire safety systems, and their effective integration into architectural design proposals, especially for multi-storeyed buildings.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Describe the need and methods of artificial/mechanical ventilation for various building types.	BT1
CO 2	Explain air conditioning processes, components, equipment, and their architectural requirements.	BT2
CO 3	Apply appropriate lift and escalator types, service criteria, and spatial requirements for integration in buildings.	BT3
CO 4	Analyze firefighting and fire safety lift for integration in buildings.	BT4

Prerequisite: Nil

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	**Artificial/Mechanical Ventilation:** Need for mechanical ventilation in buildings. Rate of ventilation for different occupancies. Methods and equipment employed for mechanical ventilation in buildings.	30	0
II	**Air Conditioning Systems:** Definition, advantages/disadvantages, basic psychrometric process, air and refrigeration cycle. Summer/winter AC, load calculations, zoning (purpose/advantages). Air distribution: ducts, outlets. AC methods & equipment: window, split, central systems. Architectural integration and equipment location in buildings	20	10
III	**Elevators and Escalators :** History and types (traction, hydraulic, etc.), advanced types (double-decker, sky lobby), lobby/interiors. Environmental considerations: location, serving floors, grouping, car dimensions, door arrangements. Service requirements: time, passenger capacity, machine room layout. Escalators: application, arrangement, spatial planning, comparison with elevators. Conveyor systems.	20	10
IV	**Firefighting and Fire Safety:** Causes, fire load, hazards, structural grading (NBC). Building classification (NBC), material behavior. Passive control: escape routes, pressurization, and compartmentation. Active systems: extinguishers, detection, alarms, sprinklers, hydrants, tanks. NBC rules for high-rise firefighting.	30	0
	TOTAL	120	

Text Book:

1. Ambrose, E.P.; *Electric Heating;* 1st Ed,; 1968; John Wiley& Sons; New York.
2. Philips, Derek; *Lighting in Architectural Design;* 1st Ed.; 1964; McGraw Hill; New York.
3. Hopkenson, R.G. & Kay, J.D.; *The lighting of Buildings;* 1st Ed.; 1969; Faber & Faber; London

Reference Books:

1. Elevators, Escalators , Moving Walkways – Manufactures catalogues
2. *National Building Code.*
3. *Electrical systems.*
4. Handbook of building Engineers in metric systems; 1968; New Delhi
5. *Principles of Refrigeration* – Roy J. Dosat
6. *Air Conditioning and Refrigeration Data Handbook* – Manohar Prasad
7. *Refrigeration and Air-Conditioning* – Don Kundwar

5th Semester		
Paper 6	**Structures V** **L-T-P-C: 3-0-0-3 Credits: 3** **Scheme of Evaluation: Theory**	**Subject Code:** ARC132G505 **Level : 300**

Course Objective: The objective of **Structures V(ARC132G505)** is to study the structural behavior and design of RCC and steel structural systems including slabs, beams, columns, footings, staircases, and water tanks, along with principles of earthquake-resistant design using relevant IS codes.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Recall design philosophies and IS code guidelines for one-way and two-way RCC slab systems.	BT1
CO 2	Understand the behavior and design process of singly and doubly reinforced beams including standard detailing as per IS 456:2000.	BT2
CO 3	Apply IS code provisions to design RCC columns and isolated footings subjected to axial and eccentric loads.	BT3
CO 4	Analyze the properties of structural steel, types of steel sections, and basic connection design using IS 800:2007.	BT4

Prerequisite:

Students must have completed **Theory of Structures – IV**, where they learned the fundamentals of RCC behavior, concrete properties, and basic Limit State Design concepts.

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	**Design Philosophies and Methods Slab Design:** • Introduction to Limit State Design; Use of IS 456:2000 • Design of One-Way RCC slabs • Design of Two-Way slabs using IS code method	12	0
II	**Beam Design:** Moment of Resistance of Beam Design of Singly Reinforced and Doubly Reinforced Beam • Design and Detailing of RCC beams having Rectangular, T and L shaped sections	12	0
III	**Column and Footing Design:** • Design of RCC columns: Axially and Eccentrically Loaded • Design and Detailing of RCC columns having Rectangular, Circular and L shaped sections	12	0
IV	**Steel Design** • Introduction to Steel: Properties and Sections • Design of simple and eccentric connections: Bolt, Rivet and Welded connections • Design of Simple column and footing connection details • Design of Tension members and Compression members in trusses	12	0
	TOTAL	48	

Text Books:

1. *Subramanian N. (2013), Design of Reinforced Concrete Structures, Oxford University Press*
2. *Varghese P.C. (2008), Limit State Design of Reinforced Concrete, PHI Learning, 2nd Ed.*
3. *S.S. Bhavikatti – Basic Structural Analysis, Vikas Publishing House*

Reference Books:

1. Ramamrutham S.; *Strength of materials*; 16th Ed.; 2009; Dhanpat Rai; New Delhi
2. IS 456:2000 – *Plain and Reinforced Concrete – Code of Practice*, BIS
3. Hibbeler R.C. – Engineering Mechanics: Statics & Dynamics, Pearson Education

5th Semester		
Paper 7	**Architectural Conservation & Documentation** **L-T-P-C: 1-0-2-3 Credits: 3** **Scheme of Evaluation: Studio**	**Subject Code:** ARC132D547 **Level : 500**

Course Objective: The objective of **Architectural Conservation & Documentation (ARC132D507)** is to introduce students to the discipline of Architectural Conservation and to develop basic skills required in documenting a Heritage structure/site.

Course Outcomes:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Identify the skills required in documenting a heritage site	BT1
CO 2	Understand the importance of Architectural Conservation.	BT2
CO 3	Develop their sense on Introduction to Architectural Conservation, General Practices	BT3
CO 4	Critically evaluate and present Conservation studio project: Research and Site Visits, Documentation and Compilation.	BT4

Prerequisite:

- Knowledge of Heritage

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	**Introduction to Architectural Conservation:** • Understanding Heritage and Documentation as a tool for recording heritage; • General Definitions • Brief History of Conservation • Types of Heritage • Tangible and intangible heritage; Traditional Knowledge system • Natural and Designed Landscape, Cultural Landscape, Sacred Landscape; • Conservation organizations in India and Abroad • UNESCO World Heritage Sites • Archaeological Survey of India (ASI) • State Archaeology • INTACH • Other related organisations	5	10
II	**General Practices:** • General practices, Theories and philosophies • Charters (Athens Charter, Venice Charter, Nara Document, Florence Charter, etc.) • Criteria and Methodology of Listing and Grading of Heritage Buildings according to ASI • Model Building Bye-laws in India • Ancient Monuments and Archaeological Sites and Remains Act (AMASR Act) –2010 • Factors deteriorating Heritage Buildings/sites	5	10

	• Conservation Technique		
III	**Introduction to Conservation studio project: Research and Site Visits:** • Introduction to Conservation studio project: Research and Site Visits • Selection of Structure/site for Documentation • Literature study	5	20
IV	**Documentation and Compilation:** • Detailed Documentation • Significance of the structure/site • Analysis and Identified issues • Proposal	5	20
	TOTAL	**80**	

Text Book:

1. *Conservation of Historic Buildings:* Feilden, Bernard M.; 3rd edition (2003), Architectural Press.

Reference Books:

1. **A History of Architectural Conservation: Jokilehto, Jukka; 2nd edition.**

6th Semester		
Paper 1	**Architectural Design VI** **L-T-P-C: 1-0-8-9 Credits: 9** **Scheme of Evaluation: Studio**	**Subject Code:** ARC132C611 **Level : 300**

Course Objective: The objective of **Architectural Design VI (ARC132C611)** is to enable the students to integrate design with services, function, vertical planning , structural design component, significance of material, construction techniques, specialized building services in the framework of architectural design , Development Regulations (building byelaws and rules); circulation networks (people, vehicular access), site planning, advanced construction technology and newer building materials and site context.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Recall the typologies, functions, and systems of multi-storeyed buildings through literature reviews and case studies.	BT1
CO 2	Understand and interpret design parameters for concept development and site planning in dense urban contexts.	BT2
CO 3	Apply principles of vertical planning, structural systems, and building services integration in detailed design.	BT3
CO 4	Analyze and produce professional architectural representations, including working drawings, 3D views, and portfolio presentations.	BT4

Prerequisite:

- **NIL**

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	Literature review and case studies: Learning from detailed study and analysis of building systems and envelopes; character of Multi-storeyed Building through literature review, site visits.	0	32
II	Concept development and site planning: Concept development of the project and detailed site planning. The studio enables understanding the complex mechanisms of designing services intensive buildings in tight urban context, having multiple levels (above and/or underground). Use of core plan, service floor, vertical stacking and vertical planning to maximise the use of concept.	0	32
III	Detailed planning: Detailed building planning with services and construction techniques considered. The special emphases are on utilitarian parameters, space optimisation, conformance with regulatory requirements, integration of structural systems and building services (HVAC, fire, electrical, communication, plumbing etc.) in architectural layout and construction technology.	0	32
IV	Representation drawings:	0	32

	Finalization of design and make presentation drawing for execution which includes Site plan, floor plans, elevation, sections, details,3D Perspectives and models.. Presentation of works of design through portfolio or panels.		
	TOTAL	128	

Remarks: Designing of Mixed-use Tower (High Rise). The studio encourages the students to explore modern automation and intelligent systems for building management and energy conservation. They will learn about site planning and landscaping in tight spatial context.

Text Books:

1. *High-rise Manual: Typology and Design, Construction and Technology* ,Kloft, E. and Johann, E. (2003)., 1st Ed. Basel : BirkhauserVerlag AG.
2. *Typology+: Innovative Residential Architecture* , Markus, K., Rollbacher, R., Herrmann, E., Wietzorrek, U. and Ebner, P. (2009. Basel: BirkhauserVerlag AG.

Reference Books:

1. Baiche, B. and Walliman, N. (2012*). Neufert Architects Data*, 4th Ed. Oxford : Wiley-Blackwell.
2. Chiara, J. D. and Michael, J. C. 2001. *Time Savers Standards for Building Types*. Singapore : McGraw Hill Professional.
3. Gauzin-Muller, D. (2002). Sustainable *Architecture and Urbanism: Concepts, Technologies, Examples*. 1 st Ed. Basel :BirkhauserVerlag AG.
4. Huxtable, A-L. (1984). *Tall Buildings Artistically Reconsidered*.

6th Semester		
Paper 2	**Building Construction and Material -VI** **L-T-P-C: 1-0-4-5 Credits: 5** **Scheme of Evaluation: Studio**	**Subject Code:** ARC132C612 **Level : 300**

Course Objective: The objective of **Building Construction and Material -VI (ARC132C612**) is to introduce students to the technicalities of building construction and material and to acquaint with various methods of construction.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	*Identify various specialized door types, external glazing, and cladding systems along with associated construction materials such as PVC and FRP.*	BT1
CO 2	Explain the function, classification, and advantages of advanced construction materials like UPVC, composite panels, and insulation systems	BT2
CO 3	Apply appropriate construction techniques for installing specialized components like collapsible gates, glazing systems, and external cladding with correct material selection.	BT3
CO 4	*Analyze the performance, specifications, and integration methods of fire, sound, and thermal insulation materials in advanced building systems.*	BT4

Prerequisite: NIL

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I A	SPECIALISED DOOR OPENINGS & GATES: Types of specialised doors- Collapsible Gate, Rolling Shutter, Steel doors for garage & Sliding gates & understanding the concept of Sensor door.	0	15
I B	Material: PVC & FRP: Study the importance of PVC & FRP Materials with respect to conventional materials, manufacturing process and its advantages in building industry.	1	0
II A	EXTERNAL CLADDING: Understanding the functions of external cladding and its importance, Types of cladding- Aluminium Composite cladding showing its fixing details.	0	15
II B	Material: SOUND & THERMAL INSULATION: Types and classification of sound and thermal insulation, properties and specifications.	1	0
III A	FRONT GLAZING: Front Glazing with glass fixing with different techniques-	0	15

	Aluminium sections, Patch and spider fittings.		
III B	Material: ADVANCED BUILDING COMPONENTS FOR SOUND & THERMAL INSULATIONS: Types and classification of Advanced building components- UPVC Doors & windows, Understanding section profiles, its advantages & specifications.	1	0
IV A	EXTERNAL GLAZING: External glazing of various types- Curtain Glazing & Structural Glazing showing with advanced fixing details.	0	15
IV B	Material: FIRE INSULATION: Types and classification of Fire insulation materials and its properties.	1	0
	TOTAL	64	

Text Books:

1. *Building construction*; Mackay, W.B.; 1st Ed.; 2005; Donheed; London.

Reference Books:

1. Chudley, R; *Construction technology*; 2nd Ed.; 1987; ELBS; Harlow.
2. Barry, R; *Construction of building*; 4th Ed.; 1999; East West Press; New Delhi.
3. Kumar, Sushil; *Building Construction*; 19th Ed.; 2001; Standard Publishers Distributors; Delhi.
4. Arora, Bindra; *Building Materials*;

6th Semester		
Paper 3	**Working Drawing** **L-T-P-C: 2-0-2-4 Credits: 4** **Scheme of Evaluation: Practical**	**Subject Code:** ARC132C613 **Level : 300**

Course Objective: The objective of **Working Drawing (ARC132C613)** is to equip students with the skills and techniques of preparing professional working drawings required for statutory approvals and on-site construction execution, through the application of architectural drafting conventions and construction documentation practices.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Identify the standard architectural drafting conventions, symbols, and office standards in drawings.	BTI & BT2
CO 2	Prepare detailed working drawings for foundation, floor plans, sections, and elevations for a given project.	BT3
CO 3	Prepare and analyze comprehensive internal service layouts including kitchen, toilets, stairs, and window-door detailing.	BT3 & BT4
CO 4	Assess and conclude coordinated service drawings like site development, water supply, electrical, and sanitation plans.	BT5

Prerequisite:

- Drawings need to be prepared for any one project handled in an earlier Architectural design studio / semester. Use of CAD techniques in working drawing may be encouraged.

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	Introduction: Architectural Drafting – Lettering, Dimensioning line, Drafting conventions Title blocks, Office standards, representation of different materials in section. Use of Graphic symbols in drawings.	10	10
II	Drawing Sheets- I: Drafting and preparing sheets of foundation plan, parking plan, floor plans, elevations, sections, roof plans with roof top water drainage layout.	6	22
III	Drawing Sheets- II: Drafting and preparing sheets of kitchen, toilet, staircase, lift, doors, and windows.	4	20
IV	Drawing Sheets-III: Drafting and preparing of sheets of site development, electrical drawings, gate with boundary wall, sanitation and water supply with calculations and detailed plans of services.	4	20
	TOTAL	96	

Text Books:

1. *Building construction*; Mackay, W.B.; 1st Ed.; 2005; Donheed; London.

Reference Books:

1. Chudley, R; *Construction technology*; 2nd Ed.; 1987; ELBS; Harlow.
2. Barry, R; *Construction of building*; 4th Ed.; 1999; East West Press; New Delhi.

6th Semester		
Paper 4	**Building Science II** **L-T-P-C: 2-0-0-2 Credits: 2** **Scheme of Evaluation: Theory + Practical**	**Subject Code:** ARC132C616 **Level : 300**

Course Objective: The objective of **Building Science II (ARC132C616)** is to develop knowledge and skills essential for understanding architectural acoustics, its principles, behavior in enclosed spaces, acoustical design integration, and noise control methods in building design.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define basic terminologies and principles related to sound and acoustics.	BT1
CO 2	Explain the behavior of sound in enclosed spaces and identify factors affecting reverberation.	BT2
CO 3	Apply acoustic principles to design requirements of halls and auditoriums.	BT3
CO 4	Analyze noise sources, classifications, and control measures in environmental and industrial contexts.	BT4

Prerequisite:

- NIL

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	Introduction to the Study of Acoustics: Nature of sound, frequency, pitch, tone, sound pressure, sound intensity, decibel scale, loudness, threshold of audibility and pain, masking, inverse square law.	7	2
II	Behavior of Sound in Enclosed Spaces: Reflection from plane/convex/concave surfaces, diffraction, absorption, sound absorption coefficient, reverberation time, Sabine's and Eyring's formulae. Sound absorbents: porous, membrane, cavity absorbers, role of functional absorbers. IS code 2526-1963, absorption coefficients of indigenous materials, market survey for acoustic products.	9	0
III	Acoustical Design : Requirements for speech halls, music halls, cinema, open-air theatres. Study of acoustically treated auditoriums. Case studies (online/offline).	4	5
IV	Environmental & Industrial Noise and Noise Control: Classification (airborne, structure-borne, impact, etc.), noise transmission and loss, permissible limits, insulation and constructional noise control, measuring instruments. Industrial sources (impact, turbulence, friction), outdoor noise (traffic, seashore). Traffic noise planning in schools, residences, hospitals.	9	0
	TOTAL	60	

Text Books:

1. *Architectural Acoustics*, David Egan; India Edition; 2008/2009; J. Ross Publishing, Florida / Cengage Learning India Private Limited; New Delhi.
2. *Acoustic Design for the Home Studio*, Mitch Gallagher; 1st Ed.; 2006; Thomson Course Technology Professional Technical Reference

Reference Books:

1. Doelle, Leslie L.; *Environmental Acoustics*; 1st Ed.; 1972; McGraw-Hill; New York
2. Knudsen, Vern O., Harris, Cyril M.; *Acoustical Designing in Architecture*; 1980; American Institute of Physics for the Acoustical Society of America; New York
3. Parkin, Peter H., Humphreys, Henry R.; *Acoustics: Noise and Building*; 1st Ed.; 1969; Faber and Faber; London

6th Semester		
Paper 6	**Universal Design** **L-T-P-C: 3-0-0-3 Credits: 3** **Scheme of Evaluation: Theory**	**Subject Code:** ARC132G604 **Level : 300**

Course Objective:

- To familiarize students with the idea of universality in design to create inclusive spaces and programs.
- To orient and encourage students to consider various stakeholders/users and their requirements in design.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	State the concept, history, and regulatory framework of Universal Design.	BT1
CO 2	Explain and analyze principles, types, and specifications of Universal Design.	BT2
CO 3	Apply real-life case examples to identify gaps and successes in inclusive design.	BT3
CO 4	Analyze design solutions incorporating Universal Design principles in practical projects.	BT4

Prerequisite:

NIL

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	**Introduction to Universal Design:** • Definition • Significance • History and Regulatory Acts (brief overview) • User participation	12	0
II	**Aspects of Universal Design** • Types • Principles of Universal Design • Specifications and standards	12	0
III	**Understanding in Practice** • Case examples of projects where universal design was not adopted – identifying gaps • Case examples of projects where universal design was adopted – analyzing user acceptability and inclusiveness	12	0
IV	**Design Tools** • Developing a design incorporating universal design principles • Projects may include: ➢ Street section ➢ Public toilet ➢ Café ➢ Bus stop ➢ Ticket counter	12	0
	TOTAL	48	

Text Books:

Preiser, Wolfgang F. E., *Universal Design Handbook*, McGraw Hill Professional, 2001.

Reference Books:

- Goldsmith, Selwyn, *Universal Design*, Taylor & Francis, 2000.
- Hamraie, Aimi, *Building Access: Universal Design and the Politics of Disability*, University of Minnesota Press, 2017.
- Vavik, Tom, *Inclusive Buildings, Products & Services: Challenges in Universal Design*, Tapir Academic Press, 2009.

6th Semester		
Paper 6	**Green Building** **L-T-P-C: 1-0-2-3 Credits: 3** **Scheme of Evaluation: Theory + Practical**	**Subject Code:** ARC132D617 **Level : 300**

Course Objective: The objective of **Green Building (ARC132D617)** is to introduce students to green concepts and techniques in the built environment, with an emphasis on sustainability, conservation of resources, and integration of rating systems to achieve environmental responsibility in architecture.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Describe the importance, need, and certification systems related to green buildings.	BT1
CO 2	Explain sustainable architecture principles with an emphasis on site selection and planning	BT2
CO 3	Apply strategies for water conservation and energy efficiency in building design	BT3
CO 4	Analyze environmental impacts of materials and propose solutions for indoor environmental quality.	BT4

Prerequisite:

- NIL

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	Introduction to Green Buildings and Rating Systems: Introduction and need for green buildings. Organizations and industries promoting green architecture. Overview of certifications: LEED, IGBC, GRIHA.	22	0
II	Sustainable Architecture and Site Selection Planning: Site preservation, soil erosion control, public transport access, topography conservation, universal design, workforce facilities, and heat island reduction.	19	5
III	Water Conservation and Energy Efficiency: Rainwater harvesting, efficient plumbing, irrigation management, wastewater treatment, energy performance and efficiency, renewable energy sources, and energy metering.	17	5
IV	Building Materials and Indoor Environmental Quality: Fresh air ventilation, tobacco smoke control, waste segregation, indoor pollutant minimization, use of green-certified materials, air quality testing and management.	17	5
	TOTAL	90	

Text Book:

1. Krishan A., Baker N., Yannas S., Szokolay S.V. (2001), *Climate Responsive Architecture*, Tata McGraw Hill Education Private Limited
2. Kordjamshidi M. (2011), *House Rating Schemes from Energy to Comfort Base*, Springer

Reference Books:

1. *IGBC Green New Buildings Rating System* (2016), Version 3.0, Indian Green Building Council

7h Semester		
Paper 1	**Architectural Design VII** **L-T-P-C: 1-0-8-9 Credits: 9** **Scheme of Evaluation: Studio/Jury**	**Subject Code: ARC132C711** **Level: 400**

Course Objective: The objective of **Architectural Design VII (ARC132C711)** is to emphasize on site planning with creative and rational skills for problem solving on real site. The design problem should focus (but not limited to) multifunctional large span public buildings in an urban setting (preferably low to medium density). Emphasis of the problem would be on design parameters (generating unique sense of architectural vocabulary), graphical representation of design details and architectural expression in functional and construction elements, energy efficient approaches etc. The design proposal will be taken up with bye-laws, city master plan or any other restriction on large sites.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Identify and form correlation between design and other subjects studied in previous semesters and till present.	BT1
CO 2	Translate ideas through sketches, drawings or models developed to meet various requirements.	BT2
CO 3	Demonstrate creative use of appropriate technology, structural system and materials.	BT3
CO 4	Analyze and integrate concerns of structure, construction, and building services into architectural drawings by critically evaluating building bye-laws, sustainable technologies, and energy efficiency parameters.	BT4

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	Introduction – Literature review, Case studies & Site visit: Develop understanding through literature review (group or individual), case studies, evaluating real-life projects, through documentaries/ movies/ photos etc. Socio-cultural approach must be sensitised to deal with such design projects. Site visits for collection of context specific data, following mapping, analysis/ evaluation.	8	24
II	Concept Development: Discussions on key-words, concept sketches initiated for idea generation. Flow diagram to explore relation of various spaces, bubble diagram for locating various zones on site, try and re-create to analyse spaces in all dimensions through block models, single line graphics and study models for choosing the right option.	8	24
III	Design Development & Area Programming: Understand the role of existing site features & built environment to design spaces of learning and recreation through formal & informal means of interactions. Consider building bye-laws & other site restrictions. Get intended outcomes through architectural design integrated into the urban fabric. The planning worked out in two stages; site plan/ master plan and detailed	8	24

	designs of individual buildings. All interior and exterior built-spaces must be planned to provide ease of access to all (universal design).		
IV	Final Design Proposal: Final set of drawings (site/ master plan, site sections, floor plans, elevation, sections, details, 3D Perspectives) and detailed physical model explaining the approach and consideration of natural & urban setting to achieve the requirements (with various other restrictions), use of innovative materials, techniques and integration of services in site as well as building level. Presenting the works of design through portfolio or panels.	8	24
	TOTAL	128	

Remarks:

Design exercises could be Performing Arts Centre/ Institute, Regional Cultural Complex, Sports Hub, Interpretation Centre, Auditorium & Exhibition Complex.

Text Books:

1. Mildred F. Schmertz, AIA. (1973). *Campus Planning and Design*. McGraw-Hill Inc.,US.
2. Dober, P. (1996). *Campus Architecture: Building in the Groves of Academe.* McGraw-Hill Inc.,US.
3. Gauzin-Muller, D. (2002). *Sustainable Architecture and Urbanism: Concepts, Technologies, Examples*. 1st Ed. Basel: BirkhauserVerlag AG.
4. Bell, Simon. (1997). *Design for Outdoor Recreation.* Spon Press 11 New Fetter Lane, London.

Reference Books:

1. Baiche, B. and Walliman, N. (2012*). Neufert Architects Data,* 4th Ed. Oxford : Wiley-Blackwell.
2. Chiara, J. D. and Michael, J. C. 2001. *Time Savers Standards for Building Types*. Singapore : McGraw Hill Professional.
3. Harris W. Charles, Dines T. Nicholas (1998), *Time Savers Standards for Landscape Architecture: Design & Construction Data.* McGraw Hill Inc.

7th Semester		
Paper 2	**Town Planning** **L-T-P-C: 2-0-0-2 Credits: 3** **Scheme of Evaluation: Theory**	**Subject Code:** ARC132C702 **Level : 400**

Course objective: The objective of **Town Planning (ARC132C702)** is to enable the students to understand the various elements, classifications and typology of human settlements and process of evolution of cities and familiarize them with Planning concepts and process in Urban and Regional Planning.

Course outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO1	Define key terms and concepts related to town planning and recognize significant historical milestones in the evolution of human settlements.	BT1
CO2	Explain the principles of urban and regional planning and describe the characteristics of urban and rural settlements as well as major historical planning approaches.	BT2
CO3	Apply land use classification systems and planning terminologies to interpret development plans such as master plans and city development plans.	BT3
CO4	Analyse the impacts of informal growth, industrialization, and urbanization on the spatial and functional patterns of cities using relevant models and planning frameworks.	BT4

Prerequisites: NIL

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/Hours***	
		L	**P**
I.	Introduction to Town Planning and Design of Cities: a) Definitions related to Planning: Urban, Rural, Town, City, Village, Urban Planning, Urban Agglomeration, Urban Sprawl, Rural-Urban Fringe, Central Business District (CBD), Policies, Acts, Schemes etc; Levels of planning, scope and components, types of planning, elements and scope, characters of a town, census definition of urban area; Goals and objectives of planning; Benefits of planning; Densities of town and constituents of town/city; Understanding the limitations of Planning; Arguments for and against planning; Role of Urban Local Bodies (ULBs) in town planning, Industrial Revolution and its impact on cities. b) Human Settlements – Urban and Rural settlements; Evolution and growth of settlements; Characteristic features of urban and rural settlements.	8	0
II.	Town Planning in Ancient and Medieval India: Cities of Indus Valley and Vedic Period, Cities of Mughal & British period; Town planning as per Vaastu Shastra.	8	0
III.	Urban and Regional Planning Concepts: a) Evolution of Planning concepts: Garden City Movement, Three Magnet Diagram, Letchworth, Welwyn and Radburn garden city concept; Planning concepts by Le Corbusier: Chandigarh City Planning; Sir Patrick Geddes: Conurbation; Doxiadis: Ekistics. b) Urbanization, Industrialization and Urban Growth, Growth pattern	16	0

	of towns/cities, Informal Growth and its impact on the city, Urban Growth Models including Concentric Zone Model; Sector Model and Multiple Nuclei Model, Types of surveys in planning. c) Regional Planning - Need, Importance & Implementation; Difference between Urban and Regional Planning; Types of Regions, Hierarchy of Planning Regions etc.		
IV	Development Plans and Development Control Regulations: a) Land use planning and classification for Urban and Rural Settlements, Importance of Colour Codes; Concept of Local Area Development Plan, Zonal Plan, City Development Plan, Master plan, Regional Plan etc. its elements, preparation and implementation; URDPFI Guidelines, DCR, Zoning Regulations, Model Building Bye laws, National Building Codes etc. b) Planning for Special Areas (hilly areas, coastal areas, border areas etc.): Guidelines, Regulations and Policy frameworks; Satellite City: Meaning, Characteristic & Examples; Smart City Planning.	16	0
TOTAL		48	

Text Book:

1. *An Introduction to the Science of Human Settlements*, C.L.Doxiadis, 1968; Ekistics Hutchinson, London.
2. *Urban Planning: Theory and Practice*, Rao M.P., 2009; Cbs Publication.
3. *The Urban Pattern City Planning and Design*, Gallion A.B, 2005; CBS Publishers and Distributors.

Reference Books:

1. Lewis, M. (1968). *The City in History: Its Origins, Its Transformations, and Its Prospects.* Mariner Books.
2. Peter, H. (2014). *Cities of Tomorrow: An Intellectual History of Urban Planning and Design Since 1880* (Fourth ed.). Wiley-Blackwell.

7th Semester		
Paper 3	**Urban Design** **L-T-P-C: 2-0-1-3 Credits: 3** **Scheme of Evaluation: Theory + Practical**	**Subject Code:** ARC132C713 **Level: 400**

Course Objective: The objective of **Urban Design (ARC132C713)** is to introduce Urban Design as a professional discipline at the intersection of architecture, landscape architecture, and urban planning, while sensitizing students to the concept of the public realm, the city as a dynamic three-dimensional entity, and the perception of spaces at multiple scales, along with an understanding of its historical evolution and foundational theories.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define the fundamental vocabulary, principles, and historical evolution of Urban Design, including distinctions between public, private, and transitional spaces.	BT1
CO 2	Describe the morphological and perceptual dimensions of urban environments through concepts like land use patterns, sense of place, and visual identity.	BT2
CO 3	Interpret the social, visual, and functional components of public spaces by conducting observations and mapping activities in urban areas.	BT3
CO 4	Analyze complex urban issues related to sustainability, governance, and informality to assess the relationship between buildings and broader urban contexts.	BT4

Prerequisite:

- NIL

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	Introduction and Fundamentals: • Brief review of the evolution of urban design as a discipline through history. • Basic principles and theories of Urban Design/Concepts • Concepts related to private, public and the interfaces. • Vocabulary and the elements of Urban Design. Determinants of urban form and· Components of urban structure.	10	06
II	The Morphological and Perceptual Dimension: • Morphological Dimension: Key Concepts - Land use, Built Structures, The Public Space Network; Traditional Urban Space; Urban block Patterns and Road Networks. • Perceptual Dimension: Human sensory perception of environment; Meaning and symbolism in urban form; Sense of Place and Placelessness; Place Identity; Key Attributes of Successful places; Invented places and	10	06

	Superficiality.		
III	The Social, Visual and Functional Dimension: • Social Dimension: Social construction of space; Safety and Security; The idea of being inclusive: Accessibility. • Elements of Visual Dimension: Patterns and Aesthetic Order; Positive and Negative Space; Streets and Squares; Townscape and Urban Architecture; Hard and soft Landscaping; Street Furniture; Activity mapping Functional Dimension: Comfort, Relaxation, Passive & Active Engagement; Movement; Privacy	6	10
IV	Urban Design and Sustainability: • Relationship of urban design with economic, environmental, and social sustainability. Influence of politics and governance in public space design. Contested spaces. (Conceptual understanding). • Concepts of formal and informal sector in public realm. Future of the city, community, and participatory methods as a way forward	6	10
	TOTAL	64	

Text Books:

1. Larice, M. and Macdonald, E. Ed. (2013). "The Urban Design Reader". 2nd Edition. The Routledge Urban Reader Series, Abingdon, Oxon: Routledge.
2. Watson, D., Plattus, A. and Shibley, R. (2003). "*Time-Saver standards for urban design*". New York: McGraw Hill.
3. Spreiregen, Paul D. (1965). "*Urban design, the architecture of towns and cities".* Mcgraw-hill Inc, USA.

Reference Books:

1. Carmona, M., Heath, T., Oc, T. and Tiesdell, S. (2010). "*Public Places Urban Spaces*". Oxford: Architectural Press.
2. Marshall, S. (2009). "*Cities design and evolution*". New York: Routledge.
3. Lang, J. T. (2005). "*Urban Design: A Typology of Procedures and Products"*. Oxford: Elsevier/Architectural Press.
4. Moughtin, C., Cuesta, R., Sarris, C. and Signoretta, P. (2003). "*Urban Design - Methods and Techniques*". Oxford: Architectural Press.
5. Jacobs, Jane. (1992). "*The death and life of great American cities".* New York : Vintage Books.
6. Lynch, Kevin (1960). "*The Image of the City*". Cambridge MA: MIT Press.
7. Alexander, Christopher (1977). "*A Pattern Language*". Oxford University Press.
8. Lynch, Kevin (1981). "*The Good City Form*". Cambridge MA: MIT Press.

Reference Materials:

1. "Revisiting Urban Design as a Discipline", Journal of Urban Design, Vol. 20, Issue 6, 2015.Taylor & Francis Online
2. Urban Morphology and Place Identity", Urban Studies, Vol. 46, Issue 10, 2009. SAGE Journals
3. "Social Construction of Public Space in Urban India", Environment & Urbanization Asia, Vol. 9, Issue 1, 2018 SAGE Journals
4. "Contested Public Spaces and Urban Sustainability", Journal of Urbanism, Vol. 8, Issue 3, 2015 Taylor & Francis Online

7th Semester		
Paper 2	**Disaster Mitigation and Management** **L-T-P-C: 1-0-1-2 Credits: 2** **Scheme of Evaluation: Theory Practical**	**Subject Code: ARC132C704** **Level : 500**

Course Objective: The objective of **Disaster Mitigation and Management (ARC132C002)** is to equip students with practical knowledge and skills in estimation, costing, specification writing, and rate analysis necessary for calculating the cost and quantities involved in the execution of various types of buildings and architectural projects.

Course Outcomes:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define and explain key concepts such as hazard, risk, vulnerability, and coping capacity in the context of disaster events.	**BT 1 / BT 2**
CO 2	Analyze disaster management frameworks and practices, including legal, strategic, and financial aspects.	**BT 4**
CO 3	Apply disaster mitigation strategies such as evacuation protocols, damage assessment, and reconstruction planning.	**BT 3**
CO 4	Evaluate post-disaster recovery efforts in relation to sustainable development goals (SDGs) and global frameworks.	**BT 5**

Prerequisite:

- **Nil**

Detailed Syllabus:

Modules	Topics (if applicable) & Course Contents	Periods/ Hours	
		L	***P***
I	**Introduction:** Disaster, Risk, Vulnerability, Hazard, Exposure, Coping Capacity, Damage. Hazards: Earthquakes, floods, cyclones, and landslides, Biohazard, Avoidable Death, Relation to natural hazard.	**8**	**0**
II	**Terminology:** Disaster Cycle, Disaster Management, Disaster Mitigation, Disaster Risk Reduction. Introduction to Disaster Management: Understand disaster hazards and how they pose disaster threats. Disaster Management Principles and Practices, Economic and Financial Aspect of Disaster Management, Strategic Disaster Management, Information Technology in Disaster Management.	**8**	**0**
III	**Disaster Mitigation:** Warning and evacuation do's and don'ts about disaster, damage survey for designing aid package and detailed survey for reconstruction and repair. Disaster Management Act: Disaster management policy; duties of professionals, disaster response policy.	**8**	**0**

IV	**Sustainable Development:** Post Disaster Reconstruction and recovery for sustainable development, issues and policies. SDGs, GAR, SFDRR, GPDRR, APMCDRR, Hyogo Framework. Case Studies on Climate Change Action: Climate Change induced disaster.	**24**	**0**
	TOTAL	**48**	

Text Books:

1. Natural Hazard and Disasters, Hyndman D. and Hyndman D., Brooks/Cole (2006)
2. Natural Hazards, Bryant E., Cambridge University Press (2005)
3. Disaster Management: A Disaster Manager's Handbook, Carter, W.N., Manila, ADB. (2006)
4. Mitigation of Natural hazards and Disasters: International perspective. Haque, C. Emdad, Springer, Dordrecht. (2005)

Reference Books:

1. Neufert, Peter; Neufert’s architects' data; 4th Ed.; 2012; John Wiley; New Delhi
2. Ramsay and Sleeper; Architectural graphic standard; 11th Ed.; 2008; John Wiley; New York.

Reference Materials:

N/A

7th Semester		
Paper 5	**Advanced Construction and Services** **L-T-P-C: 1-0-4-5 Credits: 5** **Scheme of Evaluation: Studio**	**Subject Code:** ARC132G719 **Level : 400**

Course Objective: The objective of **Advanced Construction and Services (ARC132G719)** is to introduce students to advanced construction systems, structural roofing solutions, and innovative building materials with an emphasis on constructional design principles, physical and technical considerations, and surface modification technologies.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Identify and explain advanced construction materials and techniques used in modern roofing and truss systems.	BT 1 / BT 2
CO 2	Analyze and apply construction principles in the design of complex roof structures like domes, space frames, etc.	BT 3 / BT 4
CO 3	Demonstrate knowledge of roof-fixing materials, waterproofing technologies, and advanced construction chemicals.	BT 2 / BT 3
CO 4	Integrate knowledge of construction systems to evaluate and propose suitable solutions for roofing and cladding.	BT 4 / BT 5

Prerequisite:

- Nil

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***S***
I	Advanced Trusses: Steel trusses for various spans, ridged truss, saw tooth truss with lattice girders and fixing.	4	16
II	Advanced Roofing: i) Pneumatic structure, pre-engineering metal building, metal cladding (profiled MS sheet cladding). ii) Shell roof, folded plate, geodesic domes, space frame and tensile structure.	6	24
III	Advanced Roof Fixing And Roofing Materials: Roof lighting and use of plastic, reinforced plastic, polycarbonates, asbestos, rubber and cement products.	4	16
IV	Water Proofing: Water proofing elements, construction chemicals and additives, adhesives, plaster of Paris, gypsum, polyester sealants.	2	8
	TOTAL	80	

Text Books:

1. Construction of building; 1.Barry, R; 4th Ed.; 1999; East West Press; New Delhi.

Reference Books:

1. Chudley, R; Construction technology; 2nd Ed.; 1987; ELBS; Harlow.
2. Building construction; Mackay, W.B.; 1st Ed.; 2005; Donheed; London.
3. Barry, R; Construction of building; 4th Ed.; 1999; East West Press; New Delhi.
4. Kumar, Sushil; Building Construction; 19th Ed.; 2001; Standard Publishers Distributors; Delhi.

Reference Materials:

N/A

7th Semester		
Paper 6	**Interior Design** **L-T-P-C: 1-0-4-5 Credits: 5** **Scheme of Evaluation: Studio**	**Subject Code:** ARC132D716 **Level : 400**

Course Objective: The objective of **Interior Design (ARC132D716)** is to introduce the discipline of interior design, study its principles and their application in interior design projects and foster creative ability and inculcate skills to understand and conceive architectural design.

Course Outcomes:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Identify the principles, vocabulary, and historical evolution of interior design including typologies, functions, and vernacular traditions	BT1
CO 2	Interpret and apply planning principles and interior elements like floors, walls, ceilings, and circulation in conceptual design layouts.	BT2
CO 3	*Demonstrate detailed design development including services, materials, lighting, interior landscaping, and furniture integration for a holistic interior environment.*	BT3
CO 4	Create and present comprehensive interior design solutions using detailed drawings, 3D visuals, and models that express design intent effectively.	BT4

Prerequisite: NIL

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	Introduction and History of Interior Architectural Design: Learning Definition of interior design, Interior architectural design process, design vocabulary in terms of principles and elements. Introduction to design of interior spaces w.r.t typologies, functions, themes and concepts. Brief study of history through various periods, design movements etc. Study of vernacular design of India w.r.t interior design and decoration. Introduction to design problem.	1	7
II	Design Development Stage I: Introduction to elements of interiors-floors, ceilings, walls, staircase, openings, service elements etc. and methods of their treatment by materials and methods of construction. Development of visual design idea through the exploration of layouts, concepts and plans. Planning considerations: functions, orientation, circulation, grouping, light, ventilation, privacy, climatic and ergonomic factors, aesthetics & cost.	1	11
III	Design Development Stage II: Study of Interior Lighting, Interior Landscape, other accessories	1	17

	for enhancement of interiors and their effects on space and design Values. Design Finalization-Detailed Plans, Sectional Elevations, Sketches Service (electrical, lighting, water supply, drainage, air conditioning) Materials & finishes (wood, glass, plastic, metals, acoustic boards, floor covering, panelling materials, false ceiling material) Furniture details		
IV	Presentation drawings: Presentation drawings, floor plans, sectional elevations, details, 3D Perspectives and models.	1	9
	TOTAL	48	

Remarks:

Designing interior of a building - The studio encourages the students to explore the interdependence of a building and its spaces as both of them together form an entity. It provides them with a vision that both need to be treated holistically and to be allowed to merge into each other, distinct yet together and complete.

Text Books:

1. Neufert, Peter; *Neufert's architects' data;* 4th Ed.; 2012; John Wiley; New Delhi
2. Ramsay and Sleeper; *Architectural graphic standard;* 11th Ed.; 2008; John Wiley; New York
3. Ching, F.D.K, Interior design Illustrated. New York: V.N.R. Publications

Reference Books:

1. *Time saver standards for building types*,De Chiara, Joseph and Crosbie, Michael J.;2nd Ed.; 2011; Tata McGraw Hill; New Delhi.
2. Pandya, Yatin; *Elements of space making*; 1st Ed.; 2014; Mapin Publishing; Ahmedabad.
3. Ching, F.D.K, Interior design Illustrated. New York: V.N.R. Publications

7th Semester		
Paper 7	**Research Methodology** **L-T-P/S-C –0-0-0-1 Credits: 1** **Scheme of Evaluation: Practical**	**Subject Code:** ARC132C715 **Level : 400**

Course Objective: The objective of **Research Methodology (ARC132C715)** is to introduce the students to the field of academic research and enable them to establish a strong theoretical foundation, to infuse logical reasoning to achieve clarity of thought and to orient the students to structured research in a focused manner. The process of study shall enable students to conduct in depth analysis and objective research on a topic of their interest. Students may be encouraged to select the topic which may eventually culminate in the Architectural Design Thesis.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**BT Level**
CO 1	Understand the comprehensive knowledge on the tools and methods of research and provide a way to frame the reports in a proper way.	BT1
CO 2	Outline the research and contents in a professional way. They will also secure skills required to collect, assimilate and synthesize data relevant to handling a research topic and design thesis project independently.	BT2
CO 3	Develop an understanding and secure skills required to collect, assimilate and synthesize data relevant to handling a research topic and design thesis project independently.	BT3
CO 4	Analyze the inevitable component of research that precedes any design.	BT4

Prerequisite:

- No prerequisite required

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	Introduction to Research in Architecture: • Meaning of research and Types of research: Exploratory research, Conclusive research, Empirical Research etc. • Difference between design studio, dissertation and design thesis. • Terminologies related to research: aim, objectives, scope, limitation etc. • Formulating a research: difference between research question and hypothesis: antithesis.	0	4
II	Tools and Methods: • Scientific methods of research with special emphasis on architectural research methods. • Architectural enquiry, visual observations, questionnaire	0	4

	formats of enquiry, Literature Review and case studies. • Primary and secondary data, sampling, Data analysis techniques, interpretation of data. • Introduction to Architectural criticism.		
III	Presentation Techniques: • Formats for presentation of data, case studies and analysis. • Formats for presentation of thesis design- media appropriate in the architectural profession such as two-dimensional drawing, physical models, and three-dimensional computer models.	0	4
IV	Report Writing/ Paper Writing: • Techniques in report/paper writing. • Synopsis and abstract formulation. • Presentations of relevant contextual information, interpretation of the data collected and design; reporting the design development from concept to design solution, explaining the relation of the design to existing knowledge on the topic in the form of coherently written matter. • Introduction to plagiarism: Referencing, sourcing, bibliography, annexure etc.	0	4
	TOTAL	16	

Text Books:

1. Groat, Linda N. and Wang, David, "Architectural Research Methods",2nd Edition. Publishers: John Wikey & Sons., Inc., Hoboken, New Jersey.
2. Booth, Wayne C., Colomb, Gregory G. and Williams, Joseph M. (2003), "*The Craft of Research*", Second Edition. Publishers: The University of Chicago Press, Chicago.

Reference Books:

1. Sommer, Robert & Sommer, Barbara (2002), "A Practical Guide to Behavioural Research: Tools and Techniques". New York: The Oxford University Press.
2. Creswell, John W. (2003), "*Research Design: Qualitative, Quantitative, and Mixed Method Approaches*". 2nd Edition. Thousand Oaks, CA: Sage Publications.
3. Borden, I. and Ray, K. R. (2006), "*The Dissertation: An Architecture Student's Handbook.* Second Edition. Oxford: Architectural Press.
4. Anderson, J. and Poole, M. (1998), "*Thesis and Assignment Writing*". Brisbane: John Wiley

8th Semester		
Paper 1	**Architectural Design VIII** **L-T-P/S-C: 1-0-9(S)-10 Credits: 10** **Scheme of Evaluation: Studio**	**Subject Code:** ARC132C811 **Level: 400**

Course objective: The objective of **Architectural Design VIII (ARC132C811)** is to sensitize students to the concept of the public realm, understanding the city as a three-dimensional entity, and perception of space at multiple scales; familiarizing them with implementation processes through various statutory and non-statutory guidelines.

Course outcome:

After successful completion of the course, students will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define the basic principles, evolution, and scope of urban design as a discipline intersecting architecture, landscape, and planning.	BT1
CO 2	Describe various typologies, spatial structures, and components of urban form, such as streets, squares, landmarks, and the public realm.	BT2
CO 3	Apply urban design principles and sustainability concepts to analyze real-world case studies and urban scenarios	BT3
CO 4	Design a large campus or urban precinct that accommodates multiple user groups, taking into account urban design principles and contextual sensitivity.	BT6

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***PeriodsHours***	
		L	***P***
I	Introduction, literature review and scope: Relationship between Architecture, Urban Design and Urban Planning; Brief review of the evolution of the urban design as a discipline, basic principles and theories. Broad understanding of urban forms and spaces at various spatial scales through examples from historic cities.	6	26
II	Typologies and Procedures: Concepts of public and private realm; understanding different types and procedures of urban design interventions their scale relationships; constraints and challenges of urban design in democratic versus authoritarian settings.	4	28
III	Elements of Urban Design and Urban Design and Sustainability: Understanding the city as a three dimensional element; Urban form as determined by interplay of masses, voids, order, scale, harmony, symmetry, colour and texture; Organization of spaces and their articulation in the form of squares, streets, vistas and focal points; Concept of public open space; Image of the city and its components such as edges, paths, landmarks, street features. Sustainability concept; Relationship of urban design with economic, environmental and social sustainability; Urban renewal and urban sprawl; Concepts of Transit Oriented Development, Compact City, Healthy City and Walkable City.		32

IV	Urban Design Implementation: Urban design and its control; Institutional arrangements for design and planning, their roles, powers and limitations; Types of planning instruments, structure plans, master plans and local area plans and zoning guidelines; Design communication and role of public participation.		32
TOTAL		128	

Remarks:

Design exercise could be any medium to large-scale project in the public domain, situated within an existing (and preferably compact) urban fabric, such as: redevelopment of commercial areas, waterfront development, transit hubs, market squares, densification along transit corridors, mixed-use complexes.

Text Books:

1. The Urban Design Reader, Larice, M. and Macdonald, E. Ed. (2013). 2nd Ed. The Routledge Urban Reader Series, Abingdon, Oxon : Routledge.

Reference Books:

1. Carmona, M., Heath, T., Oc, T. and Tiesdell, S. (2010). Public Places Urban Spaces. Oxford : Architectural Press.
2. Marshall, S. (2009). Cities Design And Evolution. New York : Routledge.
3. Lang, J. T. (2005). Urban Design: A Typology of Procedures and Products. Oxford : Elsevier/Architectural Press.
4. Moughtin, C., Cuesta, R., Sarris, C. and Signoretta, P. (2003). Urban Design - Methods and Techniques. Oxford: Architectural Press.
5. Watson, D., Plattus, A. and Shibley, R. (2003). Time-Saver standards for urban design. New York : McGraw Hill.

Reference Materials:

1. MoUD (Ministry of Urban Development), Government of India
2. Urban and Regional Development Plans Formulation and Implementation (URDPFI) Guidelines
3. Smart Cities Mission Guidelines
4. National Urban Transport Policy
5. NIUA (National Institute of Urban Affairs)
6. Research papers and reports on Transit-Oriented Development (TOD), Public Realm, Urban Governance.
7. Journal of Urban Design (Routledge)
8. Peer-reviewed articles on public space design, participatory planning, and sustainable urbanism.
9. Marg Publications – Urban India Issues
10. Special volumes on Indian cities and their transformations in response to socio-political and design shifts.

8th Semester		
Paper 2	**Estimation and Costing** **L-T-P-C: 3-0-0-3 Credits: 3** **Scheme of Evaluation: Theory**	**Subject Code:** ARC132C805 **Level : 400**

Course objective: The objective of **Estimation and Costing (ARC132C805)** is to equip students with practical knowledge and skills in estimation, costing, specification writing, and rate analysis necessary for calculating the cost and quantities involved in the execution of various types of buildings and architectural projects.

Course outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Define and explain the types, importance, and methods of estimation used in architectural projects.	BT 1 / BT 2
CO 2	Apply approximate estimation techniques using standard rates and identify factors influencing cost calculations.	BT 3
CO 3	Analyze and prepare rate analysis and specifications for various building components and materials.	BT 3 / BT4
CO 4	Calculate detailed estimates for load-bearing and RCC structures using MB, abstract sheets, and rate schedules.	BT 3 / BT 5

Prerequisite:

- Nil

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	Introduction: Importance of estimation, types of estimates, mode of measurement etc. Study of local SR rates, market rates, measurement book (MB), RA bill. Interim and final certificate.	8	0
II	Approximate Estimation: Load bearing and RCC framed structure cost by using PWD rates to find approximate estimate. Definition, Purpose, Types of approx. estimate and the various factors affecting each type.	8	0
III	Rate Analysis and Specifications: (a)Detailed rate analysis of building items as per current schedule of rates (CSR) of local PWD. Percentage of various materials used in buildings items like cement, steel, rubble, metal, sand brick tiles etc. (b)Specification and its types. Abstract and detailed specification for various materials and items of work used in building.	8	0
IV	Detailed Estimation: Load bearing and RCC framed structure cost using measurement book, abstract sheet, PWD rates to find detailed estimation. Definition, Types of detailed estimate. Estimation problem solving: To find out estimates (problem sum).	24	0
	TOTAL	48	

Text Books:

1. Estimating and Costing, Dutta S K; 26th Revised edition; UBSPD Publisher.
2. Estimating; Rangwala. SC; Charotar Publication.

Reference Books:

3. Neufert, Peter; Neufert’s architects' data; 4th Ed.; 2012; John Wiley; New Delhi
4. Ramsay and Sleeper; Architectural graphic standard; 11th Ed.; 2008; John Wiley; New York

Reference Materials:

N/A

8th Semester		
Paper 3	**Contemporary Architecture** **L-T-P/S-C –3-0-0-3 Credits: 3** **Scheme of Evaluation: Theory**	**Subject Code:** ARC132G807 **Level : 400**

Course Objective: The objective of **Contemporary Architecture (ARC132G807)** is to provide an understanding and appreciation of Contemporary Architecture through the various Art and Architectural movement, theory, works of master architects according to the timeline, globally and in India.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**BT Level**
CO 1	Understand the streamlined timeline of Contemporary Architecture	BT1
CO 2	Outline the insight and ideas of influential theorists from antiquity to the present time.	BT2
CO 3	Develop an understanding of Contemporary trends in Indian and Western Architecture in terms of ideas	BT3
CO 4	Analyze the Contemporary trends in Indian and Western Architecture in terms of ideas	BT4

Prerequisite:

- Basic knowledge of styles and famous architects and their ideologies.

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	World architecture and design history: • Pre-modern journey of art and architecture - Introduction and timeline • Reconstruction - Italy, France (unite d'habitation), Great Britain, Germany • The American Masters - Walter Gropius, Eric Mendelsohn, Ludwig Mies Van der Rohe, Frank Lloyd Wright • The European Masters - Le Corbusier, Alvar Aalto	0	4
II	Industrial revolution and rise in modernism: • Modern Movement - the beginning - Introduction to technology, materials, Art style, Social context (Post-Renaissance art and technology), evolution of new school school of thoughts post industrial revolution (for machine movement and against the machine movement) • Art deco, Bauhaus, Internationalism • Evolution and timeline of different phases of modern Art and architecture - De Siltj, Chicago school, Prairie school Expressionism, Brutalism, Metabolism • Skyscrapers in the United States - Alcoa headquarters, Lever House, John Hancock tower • Utopias and grand structures - Richard Buckminster Fuller, Arata Isozaki, Kisho Kurokawa, Moshie Safdie, Archigram	0	4
III	Post-modernism and Architectural theories • History and introduction -Hi technology, Neo modern,	0	4

	Critical regionalism, Memphis Milano • Deconstructivism, Pop art • Globalisation -Charles Moore, Philip Johnson, Richard Rogers, Norman Foster, Renzo piano, Jean Nouvel, Santiago Calatrava, Traditional and new rationalism, Richard Meier, Zaha Hadid, Frank O Gehry, Daniel Libeskind, The minimalist Japan, Peter Zumthor		
IV	Post independence architecture in India : • Le Corbusier, Louis I Kahn, Charles Correa, Laurie Baker, Raj Rewal • 21st century architecture in India : • Geoffrey Bawa, Gautam Bhatia, Charles Benninger, Sanjay Puri, Brinda Somaya • Architectural Criticism- Definition, Source, types of Criticism according to Wayne Attoe.	0	4
	TOTAL	16	

Text Books:

1.Contemporary Architects, Ann Lee Morgan, 2nd Ed, St James Press, London

2.Creating Architectural Theory, Jon T Lang, Van Nostrand Reinhold Company, 1987

3. Story of Contemporary Architecture, Paolo Favole, Prestel Publishing, 2011

Reference Books:

1. Geoffrey Broadbent, Design in architecture, John Wiley &Sons, 1973

2.Sarbjit Bahga and Surinder Bahga, Modern Architecture in India, Galgotia Pub. Co, 1993

8th Semester		
Paper 4	**Dissertation** **L-T-P-C: 0-0-4-4 Credits: 4** **Scheme of Evaluation: Practical**	**Subject Code:** ARC132C812 **Level : 400**

Course Objective: The objective of **Dissertation (ARC132C812)** is to understand the process of selection and developing of a thesis / dissertation project and identify the topic which best suits the student for moving further to Thesis.

Course Outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**BT Level**
CO1	Outline key principles and components of research methodology to support initial topic selection and justification for thesis-level research.	BT1
CO2	Understand and explain the rationale behind the refinement of research topics, formulation of research gaps, and the need for specific methodological approaches in architectural research.	BT2
CO3	Develop a structured synopsis including aim, objectives, scope, limitations, hypothesis, and methodology based on precedent studies and literature reviews.	BT3
CO4	Analyse the chosen research topic to formulate a coherent strategy, execution plan, and report that bridges research with the design process.	BT4

Prerequisites: Research Methodology

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods***	
		T	**P**
I	Introduction: (Continuation from writing up a research paper/report in Research Methodology) • Selection of most probable thesis topics the students want to work on with logical reasoning based on their knowledge, external information, personal experience etc. • Justification: Implementing knowledge of research methodology.	2	6
II	Detailing: • Narrowing down to the final set of topics from module I based on thorough analysis. • Formulating the goals: Detailing out the research gaps and the type of approach required to take the selected topics forward.	2	6
III	Contents: • Develop the synopsis: aim, objective, limitations, scope, hypothesis, desired outcome and possible precedent studies (including literature studies and case studies). • Structuring the Methodology.	2	6
IV	Final Report: • Prepare the reports for the research that precedes the design process for the set of topics chosen in Module II. • Prepare the strategy and execution plan for the design process.	2	6
TOTAL		32	

Remarks:

- This will increase the student's knowledge on research for a topic of interest.
- Each student will be allotted a guide by the end of the semester depending on their finalized topic to carry forward their thesis.

Reference Books:

1. Neufert, Peter; *Neufert's architects' data;* 4th Ed.; 2012; John Wiley; New Delhi
2. Ramsay and Sleeper; *Architectural graphic standard;* 11th Ed.; 2008; John Wiley; New York.

8th Semester		
Paper 5	**Professional Practice and Introduction to Constitutional Law** **L-T-P-C: 3-0-0-3 Credits: 3** **Scheme of Evaluation: Theory**	**Subject Code:** ARC132C814 **Level : 400**

Course objective: The objective of **Professional Practice and Introduction to Constitutional Law (ARC132C814)** is to familiarize students with professional procedures such as tendering, valuation, easements, and arbitration, and to introduce them to essential legal frameworks and regulations relevant to architectural practice, including constitutional law and professional ethics.

Course outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Remember the role, functions, and ethical responsibilities of architects as per the Architect's Act and COA regulations.	BT1
CO 2	Identify and describe various modes of professional practice, architectural competitions, and office structures.	BT2
CO 3	Apply knowledge of tendering, contracting procedures, and financial terms related to project execution.	BT3
CO 4	Analyze the legal aspects of arbitration, valuation, easements, and constitutional provisions relevant to architecture.	BT4

Prerequisite:

- Nil

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	Introduction to Architectural Profession: • Importance of Architecture Profession. • Difference between Profession, Trade and Business. • Architect's Act 1972. • Role of Council of Architects and Indian Institute of Architects, functions, constitutions, rules and regulations. • Code of professional conduct and Ethics and Social Responsibility.	16	0
II	Practicing Architecture and Office Structure: • Small Practice, Medium Practice and Large Practice. • Architect's role in office and site management. • Role of Contractors, Clients, Architects and Sub Consultants. • Terms and conditions of engagements, Schedule of Services. • Certificates related to buildings. • Architectural Competitions	16	0
III	Tenders and Contracts: • Terms associated with tendering. • Steps before tendering. • Notice inviting tenders, Procedure of opening and selection of tenders, Qualification Criteria, Bid selection, Cancellation of tenders. • Types of Contracts and Contract documents.	16	0

	• Knowledge about various conditions of Contract. • Earnest Money, Security Deposit, Retention Money, Mobilization Fund, Bank Guarantee, Variation and Extras, Defects after Completion, Liquidate Damage, Termination of Contracts etc.		
IV	Arbitration, Valuation and Easement: • Need and scope of arbitration, Indian Arbitration Act, Arbitrator, Umpire, Appointment, Conduct Powers, Duties, Sole/Joint Arbitrators, Procedure Award, Impeachment. • Techniques/Elements of Valuation, Factors affecting valuation, compensation, rewards etc. • Easements, Rights, types and features, Dilapidation, Insurance, Consumer Protection Act. • Byelaws, Provisions of National Building Codes. • Constitutional Laws.	16	0
	TOTAL	64	

Text Books:

1. K.G Krishnamurthy and S.V. Ravindra; Professional Practice;2014; PHI publication Pvt Ltd., New Delhi
2. Apte, V. S. (2008). Architectural Practice and Procedure. Pune : Padmaja Bhide.
3. COA ; Architects (Professional conduct) Regulations, Architectural Competition guidelines ; (1989) Council of Architecture Publications.
4. COA; Handbook of Professional Documents; (2005) Council of Architecture.

Reference Books:

1. Namavati, R. ; Professional practice; (1984) Mumbai : Lakhani Book Depot
2. Rangwala, S. C. Valuation of Real Properties. Charotar Publications.

Reference Materials:

1. Neufert, Peter; Neufert’s architects' data; 4th Ed.; 2012; John Wiley; New Delhi

8th Semester		
Paper 6	**Construction & Project Management** **L-T-P-C: 3-0-0-3 Credits: 3** **Scheme of Evaluation: Theory**	**Subject Code:** ARC132C806 **Level : 100**

Course objective: The objective of **Construction Project Management (ARC132C806)** is to provide students with an in-depth understanding of managing building construction projects through efficient utilization of resources—money, manpower, and machinery and to enable students to comprehend modern tools and methods in construction management.

Course outcome:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Recall and define fundamental concepts of project management, stakeholder roles, and organizational structures relevant to construction projects.	BT1
CO 2	Interpret planning and scheduling techniques such as Bar Charts, Milestone Charts, PERT and CPM; explain phases of project control and cost analysis.	BT2
CO 3	Apply basic principles to select and analyze the suitability and economic life of construction equipment used in real world construction scenarios.	BT3
CO 4	Analyze the process of contract management and tendering, including contract types, e-tendering systems, and standard documentation such as FIDIC.	BT4

Prerequisite:

- NIL

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	Fundamentals of Project Management: Need for Project management; Role of Stakeholders in construction projects; Organization, types of organization; Study of organizational structures suitable for building and construction projects; Qualities of an ideal construction organization; Ethics in construction industry	12	0
II	Project Planning and Controlling: Construction planning, scheduling and controlling phases; Use of management techniques- Bar Chart, Mile stone Chart; Networking using PERT and CPM; Project Cost Analysis; Risk Analysis	12	0
III	Construction Equipment: Role of equipment machinery in construction industry; Brief description of Earth moving; Transporting; Spreading; Compacting; Concreting; Pumping etc.; Various issues involved in owning, operating and maintaining of construction equipment; economic life of equipment	12	0
IV	Contract Management And Tendering Process: Contractual Approach; Contract types and documents; Pre-Contract stage; Post Contract Stage; Contract Management;	12	0

	Tender document, e-Tendering, FIDIC etc.		
	TOTAL	48	

Text Books:

1. *Gupta B.L., Gupta A. – Construction Management and Machinery, Standard Publishers Distributors, 2010*
2. *Soni S.K. – Construction Management and Equipment, S.K. Kataria & Sons, 2015*
3. *"Construction Management and Equipment", Soni S.K., (2015), S. K. Kataria & Sons*

Reference Books:

1. Gahlot P.S., Dhir B.M. – *Construction Planning and Management*, New Age International, 2008

9th Semester		
Paper 1	**Professional Training** **L-T-P-C: NA Credits: 26** **Scheme of Evaluation: Studio**	**Subject Code:** **ARC132C931** **Level : 500**

Course Objective: The objective of **Professional Training (ARC132C931)** is to provide exposure to the various dimensions of architectural practice, to make students aware with the realm of interior design discipline ranging from generation of idea, preparation of drawings to the final execution of design on site.

Course Outcomes:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Identify the codes of conduct necessary	**BT1**
CO 2	Develop skills and techniques of managing small and large scale residential and commercial interior projects.	**BT2**
CO 3	Demonstrate the importance of professional ethics and responsibility towards clients, fellow professionals, contractors, suppliers, other consultants and the society.	**BT3**
CO 4	Critically evaluate and present the final internship portfolio	**BT4**

Prerequisite:

- Knowledge of Architectural Practice

Detailed Syllabus:

Modules	*Topics (if applicable) & Course Contents*	*Periods/ Hours*	
		L	***P***
I	**Nature of Works expected to be done during training:** Exposure of the trainee to difference aspects of professional practice such as preparation of sketch designs, presentation drawings, working drawings and details, estimates, bill of quantities & Specifications. Discussions with clients, various consultants, inspection and management		
II	**Content of the Training Report:** After completion of practical training, the trainee is required to submit the following in a hard copy: • Training report should contain office profile, listing of current project being undertaken, project wise details of work undertaken by the student, trainee's own assessment and experience about office, working, projects etc. • All projects listed in the report should compulsorily correspond with the list of projects mentioned in the monthly log book. • Copies of drawing shall be attached as annex to support the content of the report. The drawing prints shall be obtained with the permission of the office and stamped/ sealed by the Head of the firm or Supervisor.	Internship of minimum 16 weeks.	
III	**Critical Analysis of a remarkable project done by the Design Firm:** Detailed study and analysis of any noted or renowned project that has been designed/ executed by the company/ firm, she / he is working for internship. This shall be presented through drawing, photographs, write ups etc.		
IV	**Detailed study of Firm's design details and construction techniques:** Detailed study of the type of integrated techniques/ materials used and incorporated as a design Statement in the projects. Samples of such details and works to be documented through drawings and on-site pictures.		

	TOTAL	**48**

Remarks

- Hands – on work and professional training in an Architect's Firm/ Office. According to the Guidelines of the
- Council of Architecture, the Architect should be a certified Licence holder.
- Students are also expected to complete site visits, attend client meetings, coordinate with different teams of the firm like contractors, site engineers etc.

Text Books:

NA

Reference Books:

NA

10th Semester		
Paper 1	**Architectural Thesis** **L-T-P-C: NA Credits: 26** **Scheme of Evaluation: Studio**	**Subject Code:** **ARC132C021** **Level : 500**

Course Objective: The objective of **Architectural Thesis (ARC132C021)** is to design a project of any scale and size (in terms of built areas) as long as the required rigor and depth is demonstrated and the national level norms are matched with. It is expected that all genres of projects (study or design) would end with a design solution and should be guided by a critical inquiry. It is expected that this project will be run as a studio with individual guidance under a project coordinator and assisted by several guides.

Course Outcomes:

After successful completion of the course, student will be able to -		
CO Level	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Identify and finalize a relevant architectural research/design topic through critical inquiry, site analysis, and case study.	**BT1**
CO 2	Develop architectural concepts into spatial and design solutions through iterative ideation, guided by site context and user requirements.	**BT2**
CO 3	Demonstrate the ability to integrate design elements, technical drawings, models, and visualizations to communicate a coherent architectural thesis.	**BT3**
CO 4	Critically evaluate and present the final architectural thesis portfolio, justifying design decisions with evidence from research, codes, and universal design principles.	**BT4**

Prerequisite:

- Research methodology and Pre-thesis

Detailed Syllabus:

Modules	***Topics (if applicable) & Course Contents***	***Periods/ Hours***	
		L	***P***
I	Introduction: Finalization of topic and submission of Case Study report on the finalized topic and discussion on requirement and site analysis.		**62**
II	Conceptualization: Development of concept, taking forward the concept into the design. Ideation and formation of spaces and designing them. Site development and block model for better understanding.		**72**
III	Mid Review: Working on site plan, site section, detailed plan, section, elevation, 3D model (physical and computerized), and report.		**72**
IV	Final Submission: 1. Report as per format (minimum 2 / 3 copies), 2. Drawing Sheets, 3. Detailed / Block Model, 4. CD (Containing- Research Paper/report from Research Methodology, Synopsis from Pre Thesis, PDF of report and drawings, Photos of Model).		**82**
	TOTAL	**288**	

Remarks

1. The case study for the required topic has to be done by the students at individual level before the start of the semester.
2. The requirements pertaining to being a universal design is compulsory.
3. At the time of the Viva examination, the student shall produce the portfolio containing the evolution of his/her design from the beginning to the final output. All the drawings and reports shall be certified by the Head of the Department as bonafide work carried out by the student during the semester.

Text Books:

Depending on the students desired topics for the thesis.

Reference Books:

1. Neufert, Peter; *Neufert's architects' data;* 4th Ed.; 2012; John Wiley; New Delhi
2. Ramsay and Sleeper; *Architectural graphic standard;* 11th Ed.; 2008; John Wiley; New York.